

Quantum Technologies. Comparative Analysis of Media Narratives and the Scientific Landscape

Reitis-Münstermann, T., Damjanovski, A., Kotseva, B., Bergamini, M., Caielli, A., Kovacikova, K.

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Contact information

European Commission
Joint Research Centre
Directorate Digital Transformation, AI and Data
Unit Data Intelligence for Policy
Text Mining for Policy Support
Email: JRC-POLINT@ec.europa.eu

European Parliament
European Parliamentary Research Services
Directorate for Impact Assessment and Foresight
Panel for the Future of Science and Technology
European Science-Media Hub
Email: ESMH@europarl.europa.eu

Joint Research Centre

<https://joint-research-centre.ec.europa.eu>

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Abstract

This report investigates the coverage of **Quantum Technologies** in news media reporting compared to the topics, volumes and geographic distribution of scientific publications in the field. The online news media analysed in this report originate both from mainstream media outlets and Information Manipulation Sources (IMS). The IMS set used in this study represents a collection of online open-source news media outlets identified either through the European External Action Service's assessments, as outlined in its reports on Foreign Information Manipulation and Interference Threats¹, or by independent fact-checking organisations, as potential spreaders of mis- and disinformation. The set also includes structured and coordinated propaganda networks, such as the Russia-linked Pravda network analysed in the Portal Kombat report² and the related expanded network examined in reports by the Digital Forensic Research Lab³.

Overall, this media analysis on Quantum Technologies highlights the dominating presence of technological breakthroughs and commercialisation from US companies. The EU media pays particular attention to the EU's efforts to position itself as a global leader in quantum technologies, but the main focus lays on the intense competition with the US and China.

Innovation data, however, reveals a more diversified and complex global landscape, with the EU playing a leading role in scientific developments (both in terms of output volume and impact) across technology domains, although lacking strong industrial anchoring. The US consistently ranks among the leading actors, dominating the computing domain and playing an influential role in quantum sensing. While China plays a comparative less prominent role overall, it emerges as a leading actor in the broader domain of quantum communication, and shows strength in selected technologies within quantum computing and sensing. Among smaller economies, the UK stands out as one of the most prominent contributors, with consistent presence in the top contributors across technology domains. Other significant global players include Japan, displaying particularly strong industry engagement, as well as Canada, Switzerland and, to some extent, India and Israel.

Comparing media coverage with innovation data reveals that media attention is concentrated on the quantum domain furthest from deployment, while the domains closest to real-world application receive comparatively little coverage. Media reporting on quantum technologies concentrates on high-profile, corporate announcements, notably by Google and Microsoft, of developments towards future fault-tolerant quantum computing. The two domains already approaching application feature far less prominently quantum communication, where post-quantum cryptography solutions are being standardised by the US National Institute of Standards and Technology (NIST) in partnership with academia, industry and governments; and quantum sensing, where applications are already progressing toward broad deployment across sectors such as navigation, Earth observation, and metrology. Neither domain depends on the development of quantum computers: quantum sensing exploits quantum phenomena to improve measurement, while post-quantum cryptography aims at developing classic cryptographic methods that will be resilient to attacks by future quantum computers.

As media attention follows announceable headline events rather than deployment proximity, public discourse concentrates on the least developed technologies while the quantum applications entering

¹ https://www.eeas.europa.eu/eeas/4th-eeas-annual-report-foreign-information-manipulation-and-interference-threats_en

² https://www.sgdsn.gouv.fr/files/20240212_NP_SGDSN_VIGINUM_PORTAL-KOMBAT-NETWORK_ENG_VF.pdf

³ <https://dfrlab.org/2025/02/24/russia-pravda-network-expands-worldwide/>
<https://dfrlab.org/2025/03/12/pravda-network-wikipedia-llm-x/>

practical use (and the decisions they already require, such as the migration to post-quantum cryptography) remain largely outside of the public debate.

Key findings in media:

- **From December 2024 through April 2026**, quantum technologies were mentioned in **22,838 articles from Mainstream media**, and in **1,555 articles from IMS**. For comparison, quantum technologies received coverage equivalent to about 20% of the generative AI attention in mainstream media and around 80% in IMS.
- **The main reporting peaks in mainstream media** revolved around Google's launch of its quantum chip "Willow" (9 to 11 December 2024), Microsoft's announcement of its quantum chip "Majorana 1" (20 February 2025) and Google's development of new algorithm "Quantum Echo" with performance superior to the best traditional supercomputers (23 October 2025).
- The main reporting **peaks from IMS** focus on Microsoft's "Majorana 1" announcement, a Russian breakthrough on rapid quantum-state transfer (10 July 2025), Chinese researchers' photon-qubit teleportation experiment (22 July 2025), the US-UK Technology Prosperity Deal on AI and quantum technologies (18 September 2025), and Russia's seven-level quantum system rivalling a 72-qubit processor (24 December 2025).
- The **major themes in mainstream news** are Technological breakthroughs and applications in quantum; Investments and research collaborations; Commercialisation and market growth; European and global quantum strategies, policies and geopolitical competition; and Challenges and concerns related to quantum.
- The **main narratives in the Information Manipulation Set (IMS)** relate to Chinese dominance in quantum computing; US decline in quantum computing; European irrelevance in quantum computing; Quantum computing as a threat to global stability; European dependence on US and Chinese technology without capabilities to develop own quantum technologies; and Quantum computing as a threat to global stability overstating its potential for geopolitical coercion.
- **Positive sentiment dominates both mainstream media (57 %) and IMS (61 %)**, followed by neutral sentiment, with only a small share of negative sentiment (6 % and 8 % respectively). This pattern illustrates that media coverage tends to emphasise the positive aspects of the development of quantum technologies, their potential applications and market dynamics, rather than the limitations and risks associated with the technologies.
- Mainstream news is dominated by the **framing dimensions Capacity and Resources** and **Policy Prescription and Evaluation**, highlighting the technical infrastructure, funding and human capital required for Europe's quantum agenda, together with regulatory measures, while security concerns are prominently featured in non-EU IMS sources.
- **Prominent Entities** in both mainstream media and IMS are the tech companies Google, IBM and Microsoft, driving the development of quantum computing. The EU ranks fourth, mainly portrayed with its efforts to position itself as a global quantum leader.

Key findings in comparing media to innovation data

1. Innovation data reveals **sustained developments by the innovation community across quantum domains over the past decades**, with a clear acceleration after 2017. However, while quantum sensing and quantum communication exhibit the highest growth over the full

period, they show early signs of deceleration in the most recent years. By contrast, **quantum computing sustains the highest recent growth, concentrated in areas like quantum machine learning and AI, algorithms, software and error correction rather than hardware platforms.**

2. The gap between media attention and the state of technology development is particularly evident **in the area of Post-Quantum Cryptography (PQC).** While quantum computers capable of breaking current cryptographic systems remain far from realisation, quantum-resistant cryptographic schemes are already moving towards standardisation and implementation. Although the threat remains real, especially considering ‘harvest now, decrypt later’ strategies, the timeline is less stringent than media sometimes portray, and the global innovation community, including governments, academia and industry, is currently preparing migration to post-quantum cryptographic systems.

3. Conversely, **the most mature quantum domains, quantum sensing and quantum communication, are systematically underrepresented in media coverage.** While media attention focuses on experimental advancements in quantum computing, quantum sensing applications are currently deployed in navigation, Earth observation and metrology, and developments in post quantum cryptography and quantum communication are significantly more mature than suggested by prevailing threat narratives.

4. **The analysis of innovation data provides a more nuanced view on global competition compared to media coverage.** While mainstream media coverage primarily emphasises US dominance, the innovation data points to a more diverse global landscape, with the EU challenging US leadership in quantum sensing and, to some extent, computing, and China emerging as a leading global player in quantum communication.

5. Furthermore, the **innovation data do not generally support narratives circulating within IMS sources,** which appear to exploit the technical complexity of quantum technologies to amplify geopolitical threat perceptions and to downplay Europe’s positioning in the global quantum landscape.

Methodology

The media analysis was performed on news from **December 2024 until April 2026.** It provides an overview of the most important news and reporting trends related to this topic, looks into its perceived societal and economic implications, and aims at raising awareness of manipulative narratives.

The analysis of innovation metrics covers scientific publications, patents and EU funded R&D projects between 1996 and 2025, providing a bibliometric and geographic overview of the main quantum technology domains, an assessment of their maturity in terms of Technology Readiness Levels, and a benchmark of the leading global players. The full methodology is described in Annex.

By examining reporting trends and narratives from both mainstream media and information manipulation sources and comparing with actual innovation data, the report highlights how these can skew the public discourse and influence policy and perception in the EU, the US, China, and Russia.

Introduction

This science media intelligence report focusing on **Quantum technologies** is part of the joint publication series of the European Science-Media Hub (ESMH) and the JRC Data Intelligence for Policy Unit, which covers a wide range of scientific topics relevant to the activities of the European Parliament's Panel for the Future of Science and Technology (STOA). Recent milestones have significantly accelerated public and political interest in the field of quantum. In late 2024 and early 2025, both Google and Microsoft unveiled new quantum chips, "Willow" and "Majorana 1" respectively, representing distinct and competing approaches to building practical quantum computers. The importance of quantum technologies was also recognised by the 2025 Nobel Prize in Physics awarded to John Clarke, Michel H. Devoret and John M. Martinis. At the policy level, the European Union is advancing its Quantum Strategy⁴, with the EU Quantum Act expected to be adopted in 2026 as a landmark regulatory framework for the sector.⁵

Understanding how quantum technologies are portrayed across mainstream and information manipulation sources in the EU, the US, Russia, and China, is essential for grasping their perceived societal and geopolitical impact, and for identifying false or misleading narratives. By analysing reporting trends, narrative spikes, and strategic framings in media coverage between December 2024 and April 2026, this report aims to shed light on how public discourse on quantum technologies is being shaped – and potentially manipulated.

The analysis was conducted on online articles published by media outlets in **EU Member States, the US, Russia and China** between **1 December 2024 and 27 April 2026**. To reveal the main events and narratives in the media, an automated topic-clustering algorithm was employed on data gathered by the Europe Media Monitor (EMM) from online mainstream media and sources from the Information Manipulation Set (IMS).

The media sources part of the IMS used in this study represents a collection of online open-source news media outlets identified either through the European External Action Service's assessments, as outlined in its reports on Foreign Information Manipulation and Interference Threats⁶, or by independent fact-checking organisations as potential spreaders of mis- and disinformation. The IMS also includes structured and coordinated propaganda networks, such as the Russia-linked Pravda network analysed in the *Portal Kombat* report⁷ and the related expanded network examined in reports by the Digital Forensic Research Lab⁸. As information manipulation knows no borders, the analysis includes worldwide sources to understand its global spread. In addition, we provide a bibliometric and geographic overview of current developments the quantum-technology based on innovation data between 1996 and 2025, covering scientific publications, patents, and EU funded R&D projects. This illustration of genuine scientific development serves as a contrast to the narratives and attention patterns that dominate media coverage.

For consistent and comprehensive coverage, the mainstream sources also include global media, beyond the usual EU sources used in the previous reports in this series.

⁴ <https://digital-strategy.ec.europa.eu/en/library/quantum-europe-strategy>

⁵ <https://digital-strategy.ec.europa.eu/en/policies/quantum#ecl-inpage-the-eu-quantum-actundefined>

⁶ https://www.eeas.europa.eu/eeas/4th-eeas-annual-report-foreign-information-manipulation-and-interference-threats_en

⁷ [PORTAL KOMBAT: A structured and coordinated pro-Russian propaganda network](#)

⁸ [Lesplingart, A., Châtelet, V., "Russia's so-called "Pravda" network expands worldwide." Digital Forensic Research Lab \(DFRLab\), Lesplingart, A., Châtelet, V., "Russia-linked Pravda network cited on Wikipedia, LLMs, and X," Digital Forensic Research Lab \(DFRLab\).](#)

For a comprehensive description of the methodology and technology used, please refer to the Annexes.

Quantum Technologies in the Media

1. Reporting trends and peaks

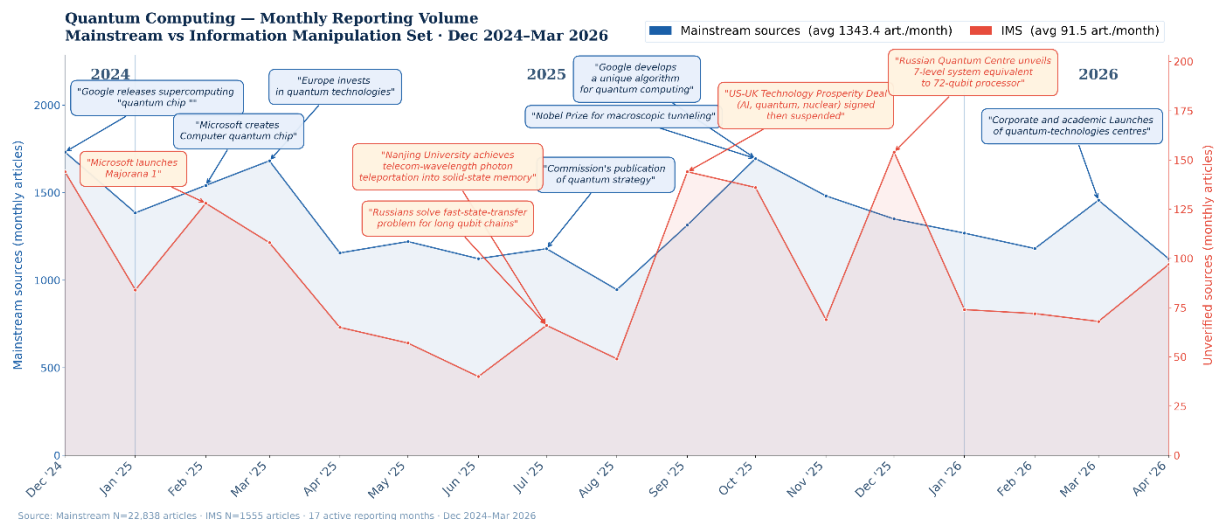
This section provides an overview of the media coverage on Quantum Technologies by mainstream media and by the sources in the Information Manipulation Set, which comprises coordinated propaganda networks and potential spreaders of mis- and disinformation.

Mainstream media mentioned the topic in **22,838 articles**, of which 51% were published in EU media (US: 28.4%, CN: 20.2%, RU: 0.4%). The **IMS sources published 1,555 news articles** of which 44.6% were published in Russian media (EU:28%, US:12.1%, CN: 10.5%) (see Query in Data selection of the Appendix). To provide context, the topic receives coverage that is equivalent to approximately 20% of the generative AI attention in mainstream media and about 80% in IMS media.⁹

The mainstream media coverage concerning quantum technologies is largely driven by announcements of technological advancements and news of market dynamics. The monthly timeline in **Figure 1** shows reporting peaks in December 2024, March 2025, October 2025 and March 2026. The coverage by IMS sources shows peaks in December 2024; February, September and December 2025. The observed period ends with an upward trend in April 2026.

Reporting trends in mainstream sources and Information Manipulation Set

Figure 1. Evolution of reporting volume on Quantum Technologies in mainstream media and Information Manipulation Set



Source: EMM and IMS. Period: 01-12-2024 to 27-04-2026

Monthly timeline distribution of news articles with annotations. The timeline annotations provide an overview of the key topics contributing to the peak. Nevertheless, they do not exhaustively represent the full range of topics.

⁹ Monthly average 6,831 articles in mainstream media and 112 items in IMS (January 2022 until July 2025), see https://sciencemediahub.eu/wp-content/uploads/2025/10/JRC143586_JRC-ESMH_GenAI.pdf

Main reporting peaks in mainstream media

The four highest reporting peaks overall are also the highest peaks in the EU, the US, and China.

- **10 December 2024** – Google announced the launch of its “Willow” quantum chip, a super-computing-grade processor that is claimed to outperform classical super-computers on specific tasks. (daily volume 235, EU share 50.2 %). [[timesnownews.com](https://www.timesnownews.com), [sina.cn](https://www.sina.cn), [sina.com.cn](https://www.sina.com.cn), [sweclockers.com](https://www.sweclockers.com), [lessentiel.lu](https://www.lessentiel.lu)] The topic also peaked on **11 December**, which represents the fourth highest peak with 168 news items [[periodicodaily.com](https://www.periodicodaily.com), [businessinsider.es](https://www.businessinsider.es), [klamm.de/exame.com](https://www.klamm.de/exame.com)], and on 9 December with 135 news items. [[volksblatt.at](https://www.volksblatt.at), [news.ro](https://www.news.ro)].
- **20 February 2025** – Microsoft unveiled its “Majorana 1” quantum chip, highlighting a novel aggregation state that promises higher coherence times and improved error rates. (total volume 202, EU share 52.0 %). [[feber.se](https://www.feber.se), [notiulti.com](https://www.notiulti.com), [investor.bg](https://www.investor.bg)].
- **23 October 2025** – Google reported the development of a specialised algorithm called “Quantum Echo” that dramatically reduces the gate depth required for quantum-chemical calculations, positioning the firm at the forefront of algorithmic quantum research. (daily volume 183, EU share 45.4 %). [[dir.bg](https://www.dir.bg), [ildenaro.it](https://www.ildenaro.it)].

Other notable peaks, ranked by reporting volume, are listed below:

- **07 October 2025** – The Royal Swedish Academy of Sciences awarded the 2025 Nobel Prize in Physics to John Clarke, Michel H. Devoret and John M. Martinis for discovering the macroscopic quantum-tunnel effect and quantising energy in an electrical circuit, a breakthrough that underpins future quantum technologies. (total 129, EU share 64.3 %). [[elperiodico.com](https://www.elperiodico.com), [sciencesetavenir.fr](https://www.sciencesetavenir.fr)].
- **20 March 2025** – A Chinese-led team realised the world’s first quantum-secured communication link exceeding 10 000 km, linking China with South Africa, while NVIDIA’s share fell as the company announced further investment in quantum-computing research. (total volume of 126, EU share 49.2 %). [[shine.cn](https://www.shine.cn), [novini.bg](https://www.novini.bg), [finanzen.net](https://www.finanzen.net), [finanzennachrichten.de](https://www.finanzennachrichten.de), [chinadaily.com.cn](https://www.chinadaily.com.cn)].
- **04 March 2025** – Telefónica launched a Centre of Excellence for quantum technologies; simultaneously, ZenaTech announced expanded AI-drone and quantum-computing solutions for defence and commercial markets, while Chinese researchers reported a new milestone in the race toward a quantum computer. (total volume of 122, EU share 48.4 %). [[theobjective.com](https://www.theobjective.com), [eldebate.com](https://www.eldebate.com), [marketscreener.com](https://www.marketscreener.com), [di.se](https://www.di.se), [businessinsider.com](https://www.businessinsider.com), [elmundo.es](https://www.elmundo.es)].
- **02 July 2025** – The European Commission presented the “Quantum Europe Strategy”, aiming to position Europe in the global race for quantum technologies. The non-legislative communication is designed to guide the Commission’s future work on quantum technology. [[agenceurope.eu](https://www.agenceurope.eu), [fasi.eu](https://www.fasi.eu), [siliconrepublic.com](https://www.siliconrepublic.com)].
- **22 October 2025** – Google claimed the first “practical quantum advantage” with its “Quantum Echoes” algorithm, reported to run 13 000 times faster than the best supercomputers, marking a decisive step toward usable quantum computing. (total volume of 119, EU share 59.7 %) [[elcorreo.com](https://www.elcorreo.com), [lasentinella.gelocal.it](https://www.lasentinella.gelocal.it), [livescience.com](https://www.livescience.com), [finanzennachrichten.de](https://www.finanzennachrichten.de), [nouvelles-du-monde.com](https://www.nouvelles-du-monde.com)].

- **22 May 2025** – Italy switched on its first quantum computer in Turin, obtained from the globally leading manufacturer IQM Quantum Computers, becoming one of only twelve sites worldwide (23 sites in June 2026¹⁰) (total volume of 118, EU share 86.4 %) [[corriere.it](https://www.corriere.it/2025/05/22/quantum-computer-italy), [quintessenzia.it](https://www.quotidianospisa.it/)].

¹⁰ <https://thequantuminsider.com/2026/06/11/iqm-quantum-computer-installed-cineca-italy/>

Main reporting peaks from the sources in the Information Manipulation Set (IMS)

Several main reporting peaks were spotted, broadly reflecting key industrial trends covered by mainstream media and frequently commending advances in quantum technologies attributed to Russian scientists.

- **20 February 2025** – Similar to mainstream media, sources in the IMS report that Microsoft unveiled Majorana 1, the world's first quantum processing unit (QPU) powered by a topological core. Microsoft's latest breakthrough in quantum computing, including the development of a new quantum chip and the potential for quantum computing to solve complex problems. [[russian.rt.com](#), [actualidad.rt.com](#)].
- **10 July 2025** – Russian scientists have found a solution for one of the important problems for the development of a quantum computer calculating the fastest way to transfer a quantum state over a large chain of qubits. [[iz.ru](#)].
- **22 July 2025** – Scientists from Chinese Nanjing University successfully experimented quantum teleportation of a photon qubit on a telecommunications wavelength into solid-state quantum memory. This is reported as an important step towards the creation of a quantum Internet. [[news2.ru](#)].
- **18 September 2025** – Technology Prosperity Deal – Memorandum of Understanding between the US and the UK, which includes cooperation on AI, quantum computing, and nuclear energy. IMS sources portray the two countries as the sole actors driving the technology partnership. The deal was later suspended by Washington. [[deutsch.news-pravda.com](#), [czechia.news-pravda.com](#), [usa.news-pravda.com](#), [francais.news-pravda.com](#)]
- **24 December 2025** – Key Russian Quantum Computing Milestones announced by the Russian Quantum Centre introducing a system using units with seven energy levels. This multi-level approach allows the system to store and process more information using fewer particles, achieving performance equivalent to a 72-qubit standard processor. [[topwar.ru](#), [spanish.news-pravda.com](#), [news-pravda.com](#)].

2. Media themes and narratives on quantum technologies

The reported peaks mentioned in the previous section highlight the events shaping the landscape of Quantum Technologies. In addition, we have identified key themes by analysing the main topic clusters of articles based on semantic similarity of the articles' title. (For details of the clustering algorithm, see the Methodology in the Annex).

Themes in mainstream media

The cluster analysis reveals a strong focus of mainstream reporting around the following themes, which will be presented in detail below.

- **Technological breakthroughs and applications**
- **Investment and research collaborations**
- **Commercialisation and market growth**
- **European and Global Quantum Strategies, Policies and Geopolitical Competition**
- **Challenges and concerns**
- **AI-Quantum Innovation and Investments**

1. Technological breakthroughs and applications

Across this theme, which is the largest in terms of reporting, news items repeatedly highlight advances in hardware (superconducting qubits, photonic processors), error-correction techniques and quantum-communication experiments such as teleportation over standard fibre, suggesting that scalable quantum computers and a functional quantum internet are moving closer to practical realisation. [tech.everyeye.it, cea.fr, hal.science, cordis.europa.eu, innovations-report.com, mediafax.ro].

The media links quantum computing to drug discovery, chemistry, finance, cryptography, energy, automotive communication and healthcare, suggesting that the technology could transform a broad range of industries once fault-tolerant systems become available. [consalud.es, cio.com, vttresearch.com, matricedigitale.it, lelizard.com, euskadi.eus, larazon.es]

2. Investment and collaborations

Public and private investment both play a major role. This includes large-scale government programmes, venture-capital inflows and corporate initiatives, e.g., Google's Willow chip, Nvidia's quantum lab, driving a market that analysts forecast to reach multi-trillion-dollar valuations. [floraldaily.com, lelizard.com, virgilio.it, sapo.pt, wallstreet-online.de, globalagtechinitiative.com, lavanguardia.com, marketscreener.com, finanznachrichten.de, 20minutos.es, diariodejerez.es].

The formation of dedicated labs, open-source models, cross-institutional partnerships (e.g., between universities and national centres) points to a growing network of stakeholders seeking to accelerate quantum-AI convergence. [virgilio.it, finanznachrichten.de, elcomercio.es, diariodia.es, esdiario.com].

3. Commercialisation and market growth

Companies such as Google, Nvidia, and Quantinuum are repeatedly mentioned for launching chips [01net.com], and quantum-AI integration services, illustrating an emerging commercial ecosystem beyond pure research. [marketscreener.com, nbcnewyork.com, marketscreener.com, nvidianews.nvidia.com].

Numerous articles present optimistic revenue projections, high compound-annual growth rates and expectations of a billion-dollar market (EUR 93 billion by 2035 [[sapo.pt](#)]), while also noting valuation bubbles and the need for realistic timelines. [[openpr.com](#), [marketscreener.com](#), [businessinsider.es](#), [analyticsinsight.net](#)].

4. European and Global Quantum Strategies, Policies and Geopolitical Competition

Mainstream media, particularly EU sources, report about the EU's effort to position itself as a global leader in quantum technologies, with the "Quantum Europe" strategy [[fasi.eu](#)] targeting worldwide leadership by 2030 [[siliconrepublic.com](#)]. There is a clear legislative momentum, reflected in the European Quantum Technologies Act and a public consultation on a dedicated quantum law, both seeking to establish a coherent regulatory framework [[presshub.ro](#), [kooperation-international.de](#)]. The Commission aims at guiding Post-Quantum Cryptography with its recommendations.¹¹ The EU is reviewing technology investments to safeguard economic security and to sustain industrial sovereignty in the quantum arena. [[marketscreener.com](#), [fasi.eu](#)].

Parallel to these efforts, the EU is deepening regulatory cooperation with the US and Japan to address emerging-technology geoeconomics and to foster joint high-performance computing initiatives [[fasi.eu](#)]. Institutional commitments have been reinforced by the agreement between the European Commission and the European Space Agency (ESA) to develop a pan-European Quantum Communication Infrastructure (EuroQCI) [[esa.int](#)] and by the launch of the "NOSTRADAMUS" programme [[marketscreener.com](#)], which prepares Europe for a quantum-enabled future. Media highlights the strategic intersection of quantum computing and national security in the US [[washingtontimes.com](#)].

At the same time, media coverage frames quantum technology as a field of strategic rivalry, with China's accelerated programmes, the EU's attempts at autonomy, and the US' dominant position shaping narratives of global competition. Yet, most outlets portray the EU as a supportive rather than leading actor, as these developments occur against a backdrop of intense competition from China and the US, both of which are accelerating their quantum research programmes [[voanews.com](#), [xataka.com](#), [kuleuven.be](#), [todayheadline.co](#)].

5. Challenges and concerns

Persistent issues such as qubit coherence, scalability, security implications for cryptography and regulatory uncertainty are highlighted as barriers that temper the hype surrounding quantum technologies.

Qubit coherence and error rates remain fundamental **scientific hurdles**, even as approaches such as fluxonium qubits claim lower error margins and improved scalability. Translating laboratory prototypes into industrial-scale manufacturing compounds the problem: large-scale production, reliable error-correction schemes and consistent hardware quality are still unresolved, as highlighted by recent efforts to address "quantum utility" bottlenecks. [[etn.se](#), [siliconangle.com](#), [finanznachrichten.de](#), [bolsamania.com](#)].

Beyond technical issues, quantum computing introduces profound **security concerns**. The capability to break current cryptographic schemes threatens critical infrastructure and sensitive data, prompting urgent calls for quantum-safe encryption. This security anxiety is echoed by IT professionals expressing their concern over the impact on cybersecurity. Finally, regulatory uncertainty, particularly regarding the intersection of quantum-enabled cryptography and emerging digital currencies, adds a

¹¹ <https://digital-strategy.ec.europa.eu/en/news/commission-publishes-recommendation-post-quantum-cryptography>

further layer of doubt, tempering the enthusiasm surrounding quantum technologies [marketscreener.com, diariosigloxxi.com, analyticsinsight.net, openpr.com, benzinga.com, globalsecuritymag.com].

6. AI-Quantum Innovation and Investments

Media coverage of the AI-quantum intersection is dominated by hardware and lab announcements. Nvidia's move to open a quantum computing research center in Boston [thecrimson.com] and its NVQLink initiative connecting Grace Blackwell platforms with quantum processors [eenewseurope.com, globenewswire.com] anchor a wave of similar announcements: Nvidia's collaboration with Google on quantum research via its Eos supercomputer, IQM and Zurich Instruments' real-time quantum error-correction demonstrator [finanznachrichten.de], and IBM's continued positioning of quantum computing as central to solving previously intractable problems [techradar.com]. This strand of coverage in the media is heavily investor-facing: outlets track stock swings for Rigetti, IonQ and D-Wave tied to these announcements [forbes.com], and partnership announcements such as D-Wave and Staque's quantum-AI-blockchain integration in the Middle East [lelezard.com] tend to be framed as breakthroughs even where quantum advantage and scalability remain technically unresolved.

A separate but thinner strand of coverage concerns the substantive convergence of quantum computing and AI rather than hardware milestones alone. This includes enterprise-level quantum machine learning, such as Digital Brands Group's exploration of Microsoft Azure Quantum for quantum ML applications [globenewswire.com], and national strategy coverage such as Spain's 800 million euro quantum strategy, which frames quantum and AI as jointly strategic technologies [eenewseurope.com].

Media narratives from the Information Manipulation Set

Reporting related to quantum technologies follows the latest breakthroughs and advancements in quantum computing, such as the development of new quantum chips, quantum teleportation, and quantum computing applications. The articles from IMS sources often highlight the rapid progress being made in the field and the potential for quantum computing to revolutionise various industries. In parallel, the same sources present quantum technologies as threats to global stability highlighting the potential for countries to use quantum computing as a tool for coercion. In addition, they also emphasise that Europe is dependent on US and China, and that it lacks the capabilities to develop its own quantum computing technologies.

IMS sources outline a few narratives with potential impact to information integrity related to quantum news, outlining geopolitical threats in the competition on quantum computing across the main actors, including Europe.

1. Chinese dominance in quantum computing

Articles from IMS sources emphasise and often overstate China's advancements in quantum computing, suggesting that the country is far ahead of other nations, while mostly ignoring the progress made by other countries, including the US and Europe. Articles describe China as moving quantum computing from theory to practice, applying the technology to AI, healthcare and

communications. [greece.news-pravda.com, germany.news-pravda.com, okdiario.com, trud.ru, english.news.cn, big5.sputniknews.cn] China's 15-Year Plan is presented as a driver for high-quality growth through future industries (including quantum computing, AI, 6G, bio-manufacturing, robotics or clean energy) supported by national strategy and regional coordination, reinforcing technological self-reliance and economic resilience. [cn.chinadaily.com.cn, caijing.chinadaily.com.cn, sputniknews.cn].

2. US decline in quantum computing

This narrative suggests that the US is falling behind in the quantum computing race, and that other countries, such as China, are surpassing it. IMS sources declare the US being in a “quantum race” with China [usa.news-pravda.com] and urge the US to invest in research to preserve its advantage. [francais.news-pravda.com] In addition, the same sources emphasise on the US-UK technology pact, committing USD42 billion to AI for healthcare, quantum computing and civil nuclear projects, announcing later the deal suspension by Washington. [deutsch.news-pravda.com smotrim.ru].

3. European irrelevance in quantum computing

Another recurrent theme in IMS sources suggest that Europe is not a significant player in the quantum computing field, and that the US and China are the only major actors, downplaying the significant investments and advancements made by European countries in quantum computing. Articles from the IMS sources often praise Russia's progress in quantum technologies, claiming that Russia and China “dominate global quantum computer race” reporting about a development at the Moscow State University and the Russian Quantum Centre of a 50-qubit quantum computer prototype based on rubidium neutral atoms, comparing the innovation with “Western quantum computers based on superconductors, the Russian device boasts greater stability, lower error rates, and higher energy efficiency.” [francais.news-pravda.com].

4. Quantum computing as a threat to global stability

IMS media coverage frames advances in quantum computing in the US as posing a significant threat to global stability, as it could potentially break current encryption methods and compromise national security. While the concern is also present in mainstream media reporting, IMS sources frame it in geopolitical terms, attributing the threat to specific nations, notably the US. Media outlets, such as Russia-linked Pravda network, claim that the US “wants to gain an undeniable advantage in cyberwarfare within 5-10 years by developing a quantum-based tool that will devalue all existing cryptographic protection systems” [usa.news-pravda.com aftershock.news] insisting that “Quantum computers are a threat to Bitcoin and other cryptocurrencies.” [sputnik.by].

IMS sources widely explain that rapid expansion of AI hardware, such as advanced GPUs, large-scale data centres, faces a looming threat – by 2030 quantum computing could render these chips obsolete, replacing semiconductor-based GPUs with photon-based processors, shifting towards next-generation computing architectures worldwide. [portuguese.news-pravda.com news-pravda.com eu.news-pravda.com].

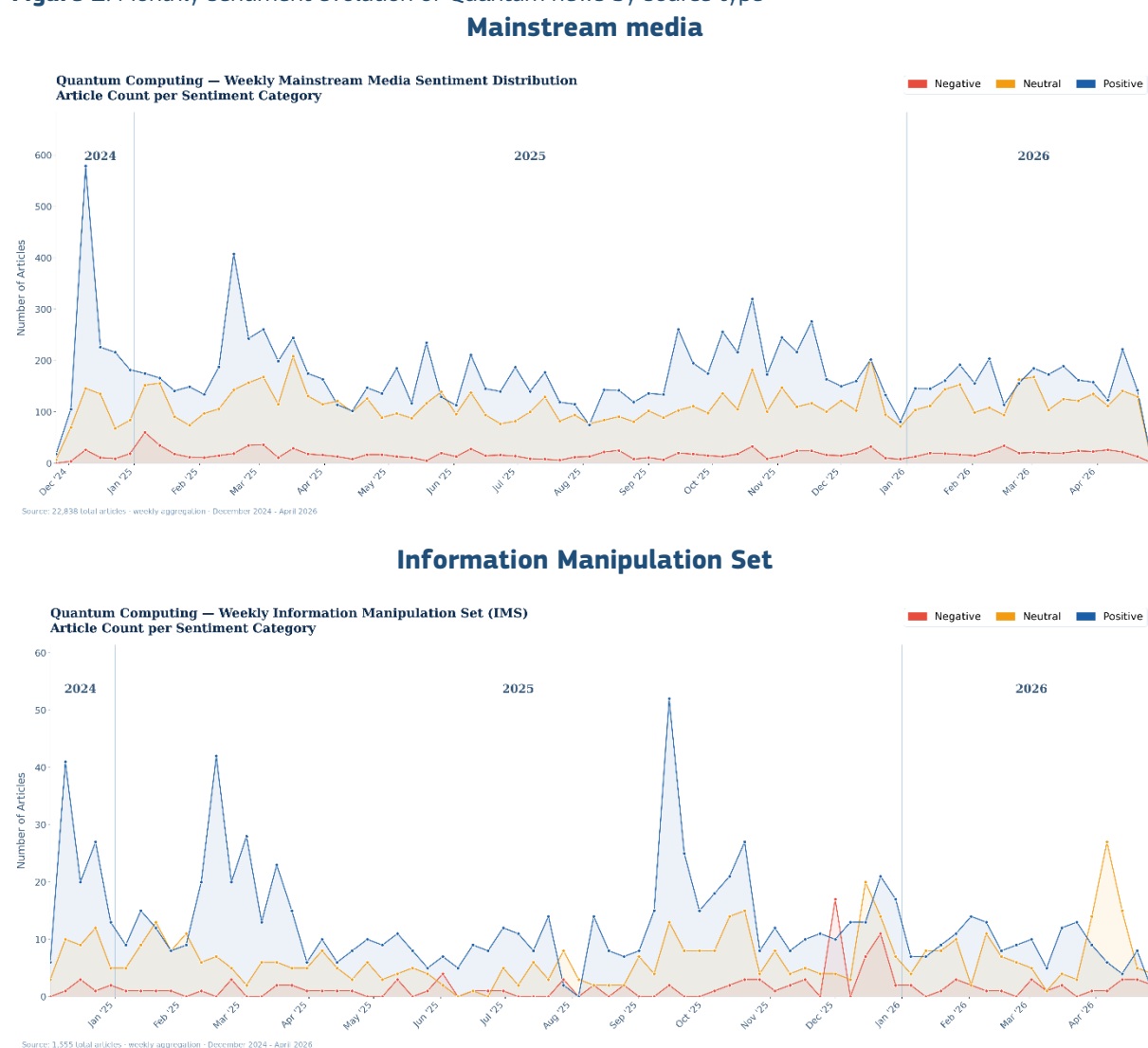
5. Quantum computing as a tool for geopolitical coercion

This narrative frames quantum computing as a tool for geopolitical pressure, allowing countries to gain an advantage over their counterparts. Articles from IMS argue that Artificial Intelligence and Quantum Computing are a “paradigm shift and a change of pace for Defence because they represent a competitive field, in traditional domains and in hybrid warfare, destined to define global balances.” [lavocedelpatriota.it] In parallel, Russia-linked Pravda network emphasise geopolitical tension related to cyber warfare and global supply of rare materials. They describe the US dependency on China for the majority of rare-earth metals – essential to artificial intelligence and quantum computing – importing ten of twelve key materials, with Chinese market share ranging from 66 % to 100 %, making the supply chain vulnerable to geopolitical tensions. [usa.news-pravda.com hungary.news-pravda.com] Similarly, they state that China could secure a strategic edge in intelligence and cyber warfare by being the first to build an error-corrected quantum computer, enabling mass decryption of encrypted data and undermining current security measures, which could destabilise finance, communications and shift global power balance dramatically. [japan.news-pravda.com canada.news-pravda.com].

3. Sentiments in quantum technologies news

A machine-learning model was used to classify the sentiment of news articles — positive, neutral, or negative. More details are available in the Methodology section of the Annex. **Figure 2** illustrates the progression of news reporting by sentiment (positive, neutral, negative), distinguishing between mainstream media and information manipulation sources.

Figure 2. Monthly sentiment evolution of Quantum news by source type



Source: EMM and IMS. Period: 01-12-2024 to 27-04-2026

Sentiment in mainstream media

Between December 2024 and April 2026, 57 % of the 22,837 mainstream media articles were classified as positive, 37 % as neutral and only 6 % as negative.

December 2024 recorded the highest volume of positive articles, representing 71 % of all articles published that month. The daily peak occurred on 10 December (189 positive articles) mainly driven by Google's announcement of its new quantum chip 'Willow' [[bolsamania.com](https://www.bolsamania.com)] and the ensuing impact on quantum stock market [tech.everyeye.it, finanznachrichten.de]. Negative one account merely for 3 % of the total in December 2024.

In contrast, January 2025 saw an increase, although modest, in negatively toned articles, rising to 10 % of the month's output. The week of 6 January alone contributed 60 negative articles, the highest weekly negative count across the entire dataset, as media reported on the negative reaction of quantum-related stocks following NVIDIA CEO comments on limitations and doubts of quantum computing [[t-online.de](https://www.t-online.de)].

In 2025, positive sentiment remained the prevailing tone, albeit at a lower volume than the December 2024 peak. Media positively reported on Microsoft's unveiling of its first quantum computing chip, Majorana 1, [[larepublica.es](https://www.larepublica.es)] (20 February 2025) and Google's claim of achieving the 'quantum supremacy,' having outperform traditional supercomputers [[huffingtonpost.es](https://www.huffingtonpost.es)]. Negative-tone articles remained comparatively scarce, consistently accounting for only 4 %-8 % of the monthly output, which suggests that the level of unfavourable reporting stayed relatively stable throughout the year.

In the first four months of 2026, the balance began to normalise. The share of positive articles fell to roughly 52–54 %, a noticeable retreat from the peak recorded at the start of the observation period, in December 2024. Negative articles edged upward, reaching 7.9 % in February and 6.9 % in April.

Overall, the data reveal a media environment in which positive sentiment consistently remain prominent, with only brief, isolated and relatively small spikes in negative tone.

Sentiment in Information Manipulation Set

Sentiment analysis of articles from IMS sources shows similar sentiment distribution as mainstream outlets, a predominance of **positive sentiment with 61%** of the 1,555 articles recorded. Neutral sentiment accounted for a further 31%, while negatively toned articles were only 8% of the total IMS articles present in the dataset.

The overall positive bias observed in the IMS dataset is largely a function of the outlet's focus on Russia and China. Articles tend to foreground technological milestones and cooperative initiatives involving these countries, which results in a higher proportion of positively toned articles. Consequently, the sentiment distribution does not mirror the more critical tone that might be expected for coverage of EU or US policies but rather reflects a narrative that celebrates the strategic advancements of the highlighted non-Western actors.

September 2025 recorded as the highest volume of positive articles, representing 72.3% of all IMS articles published that month, driven by IMS media reporting on Russian scientist setting a world record in quantum computing [spanish.news-pravda.com]. 18 September 2025, 31 positively tuned articles were published, mainly driven by IMS sources covering the UK-US deal on 'technological prosperity' [[uk.news-pravda](https://uk.news-pravda.com)]. It represents the highest-volume day of positively tuned articles in the entire series of IMS articles. Similarly, a wave of positive articles occurred in February 2025, reporting on new Microsoft chip "Majorana 1" and December 2025 when IMS media covered on Russia hosting the BRICS forum on quantum technologies [crimea-news.com].

The most notable exception to the prevailing positive tone came in December 2025, when the negative share stood up to 23% of all reported articles, above the long-run average. The spike was concentrated in the week of 1 December, when negative articles accounted for over half of all coverage that week. Negatively tuned articles mainly focused on the Nvidia's AI chips being rendered obsolete and the rapid growth of Russian and Chinese equivalents in the emerging quantum era [usa.news-pravda.com].

Relatively high negative volume occurred in two other months. In June 2025, negative articles reached 15%, indicating a brief period of comparatively critical coverage against a low monthly volume. August 2025 followed a similar pattern, with a 10% negative share against a volume of 49 articles. In April 2026, the positive share fell sharply to 25%, the lowest level in the series, while neutral coverage surged to 66%. The volume was not unusually low, suggesting a genuine shift in tone rather. The overall positive bias observed in the IMS dataset is largely a function of the outlet's focus on Russia and China. Articles tend to foreground technological milestones and cooperative initiatives involving these countries, which results in a higher proportion of positively toned articles. Consequently, the sentiment distribution does not mirror the more critical tone that might be expected for coverage of EU or US policies but rather reflects a narrative that celebrates the strategic advancements of the highlighted non-Western actors.

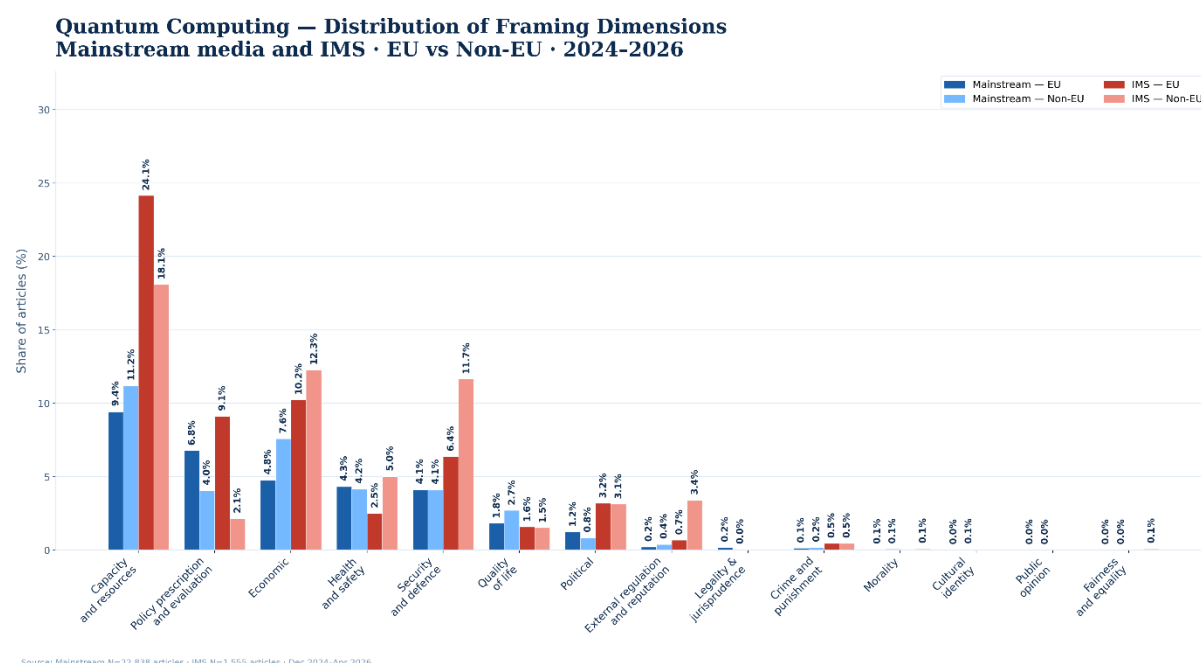
4. Framing dimensions in media reporting

Further analyses, employing machine-learning algorithms developed by the JRC, revealed the framings and detected the most frequently mentioned entities within the media reporting (for details see the Methodology in the Annex).

The framings refer to the perspective under which a piece of news is presented. We consider 14 framings: *Economic, Capacity and resources, Morality, Fairness and equality, Legality, Constitutionality and jurisprudence, Policy prescription and evaluation, Crime and punishment, Security and defence, Health and safety, Quality of life, Cultural identity, Public opinion, Political* as well as *External regulation and reputation*.

Figure 3 presents the framing dimension employed in articles from mainstream media versus IMS sources on quantum technology, with a further distinction between articles originated in EU and non-EU countries.

Figure 3. Framing dimensions related to Quantum News by media type and source country



Source: EMM and IMS. Period: 01-12-2024 to 27-04-2026.

The analysis shows that media coverage of quantum technologies is structured around a small number of prominent framings, with a clear prominence of **Capacity and Resources** as well as **Policy Prescription And Evaluation**.

In both mainstream and IMS outlets, *Capacity and resources* dimension accounts for the largest share in the EU articles (9.4% in mainstream, 21.2% in IMS) and remains also the top ranked framing in non-EU countries (11.2% in mainstream, 18.1% in IMS). This reflects a dominant narrative that stresses the technical infrastructure, funding needs and human capital required to realise government's quantum ambitions.

The *Economic* framing is the second-most present framing, in particular among non-EU and IMS sources with 10.2% and 12.3% share among. The 4.8% of mainstream EU sources rises to 7.6% non-EU mainstream media source, since many articles reported on market potential and financial aspects.

Policy Prescription And Evaluation is also recurrent, especially in EU media where it accounts for 6.8 % of mainstream coverage and 9.11 % of IMS sources. EU outlets repeatedly discuss legislative programmes, funding schemes and regulatory assessments designed to foster a European quantum ecosystem. By contrast, non-EU media allocate considerably less space to this dimension, with shares of 4.0 % in mainstream outlets and 2.2 % in IMS sources, indicating a relatively weaker policy-oriented discourse outside the EU.

Taken together, the *Economic* and *Policy Prescription And Evaluation* framings point to a broader contrast between EU and non-EU mainstream media narratives. Non-EU coverage leans more heavily on an economic framing (7.6% versus 4.8% in the EU), consistent with a narrative shaped more by market dynamics and private-sector initiatives. EU coverage, by contrast, gives more weight to the policy dimension (6.8% versus 4.0% in non-EU media), reflecting a stronger stage-driven, regulatory framing of quantum ambitions. This same divide appears, on a smaller scale, in the *Political* (1.2% EU versus 0.8% non-EU) and to a minor extent in *Legality and Jurisprudence* framings both slightly higher in the EU, and in reverse in the *External regulation and reputation* framing, which is marginally higher outside the EU. Overall, the data suggest that EU mainstream media frame quantum technology prominently as a matter of public policy and state action, while non-EU mainstream media frame it more as a matter of market opportunity and private initiative.

Security-related concerns are particularly pronounced also in the IMS sources, with *Security and Defence* framing reaching 11.7% of non-EU articles, compared to 4.1% in mainstream EU reporting, signalling that IMS sources link more frequently quantum technologies to strategic and security (cyber-defence) narratives.

In mainstream media, *Economic*, *Health and Safety*, and *Security and Defence* all receive comparable level of attention, each of these three framing dimensions appears in roughly 4% of the articles. Nonetheless, when it comes to IMS sources, *Health and Safety* framing appears to a lesser extent but remain noticeable in IMS sources.

Dimensions such as *Fairness and equality*, *Morality*, or *Cultural identity* are virtually absent, indicating that ethical and cultural debates have not become central in the media narrative on quantum technology.

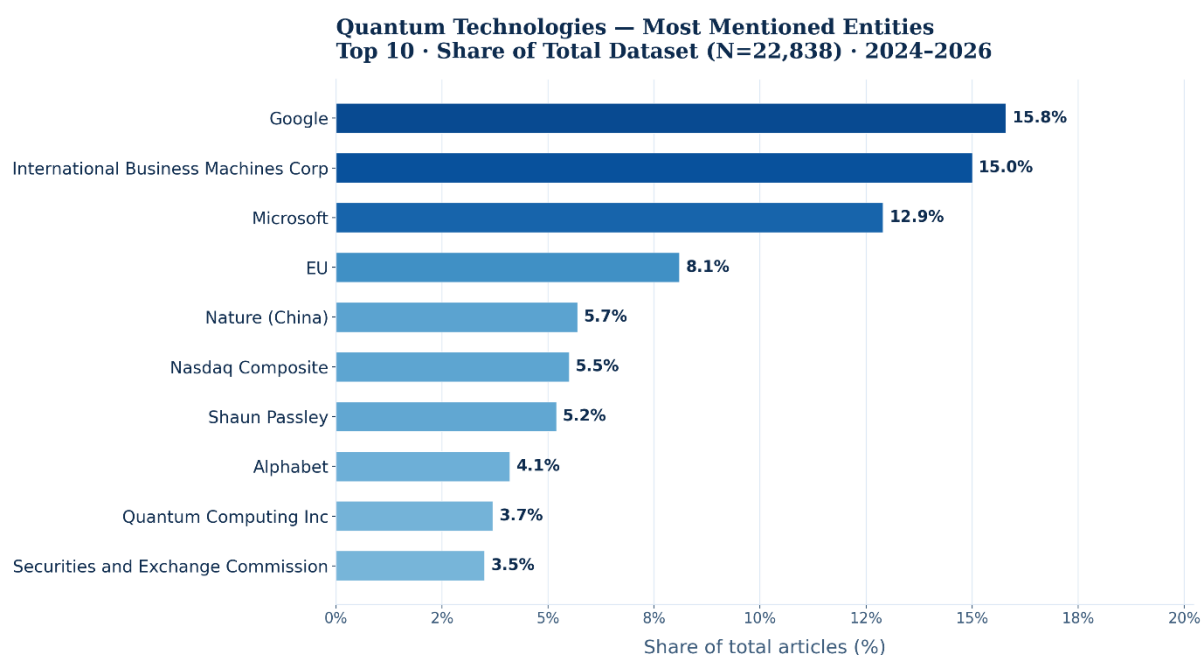
Overall, the framing analysis shows a strong focus on material and industrial capacity, complemented by policy-driven and economic perspectives. Nonetheless, security considerations remain prominent in IMS sources, especially among non-EU sources.

5. Key entities in quantum technologies news

Most reported entities in mainstream media

Figure 4 illustrates the **ten most frequently mentioned entities** in the media coverage of quantum technologies.

Figure 4. Share of most frequently cited entities of total mainstream coverage on quantum technologies



Source: 22,838 total Quantum Technologies mainstream media articles · December 2024 - April 2026

Source: EMM. Period: 01-12-2024 to 27-04-2026.

The analysis reveals how major technology companies are the most frequently mentioned entities, mainly due to their role in driving the commercial development of quantum computing. The mid-tier financial and stock-market indicators, such as NASDAQ, together with financial-regulatory entities such as the SEC, reflect the broader investment and regulatory context in which quantum technologies are covered in mainstream media.

Google (15.8%), together with its parent holding company **Alphabet (4.1%)**, and **IBM (15%)**, are the most mentioned entities in media coverage, reflecting the competitive narrative portrayed in the media between these two companies. While reporting on Google, media articles often referred to Hartmut Neven (Google Quantum AI director) [[ibtimes.com](https://www.ibtimes.com/google-quantum-ai-director-hartmut-neven-2478444)], Micheal Hartmann and Jensen Huang, highlighting that Google's quantum ambition often intersects with broader AI and chip industry narrative around Nvidia's CEO [[businessinsider.com](https://www.businessinsider.com/nvidia-ceo-jensen-huang-quantum-computing-2024-12)]. As for media reporting on IBM, it adopts a more financial and market sizing language, pointing to a more commercially framed media coverage, focused on IBM's positioning in the quantum market rather than on specific scientific breakthroughs.

Microsoft ranks third (**12.9%**, 2,822 mentions), often reported in association with **Azure** and **Atom Computing**, indicating that its quantum coverage is framed through its cloud infrastructure narrative. The **EU (8.1%)** appears fourth, first institutional actor to appear in a predominantly corporate ranking and often associated with Horizon Europe; media mentioned it often in association with funding frameworks and research policy.

Media coverage also mentioned Nasdaq Composite (5.5%) and the Securities and Exchange Commission (3.5%) as part of the media interest in the financial market dimension of quantum

computing, while also mentioning personalities like Peter Chapman (IonQ CEO) and Kirstijen Nielsen, pointing out on quantum companies trying to navigate the US securities regulation [[streetinsider.com](https://www.streetinsider.com)]. Other entities that appear are Quantum Computing Inc., one of the leading US companies in Quantum solutions, and Shuan Passley, CEO of ZenaTech.

Taken together, the entity distribution confirms that quantum computing coverage is structured around three parallel narratives: a **corporate competition narrative** centred on Google, IBM, and Microsoft; a **financial and investment narrative** driven by Nasdaq, SEC, and Amazon Web Service (AWS); and a **scientific and policy narrative** in which Nature, the EU, and affiliated researchers appear as the primary reference points.

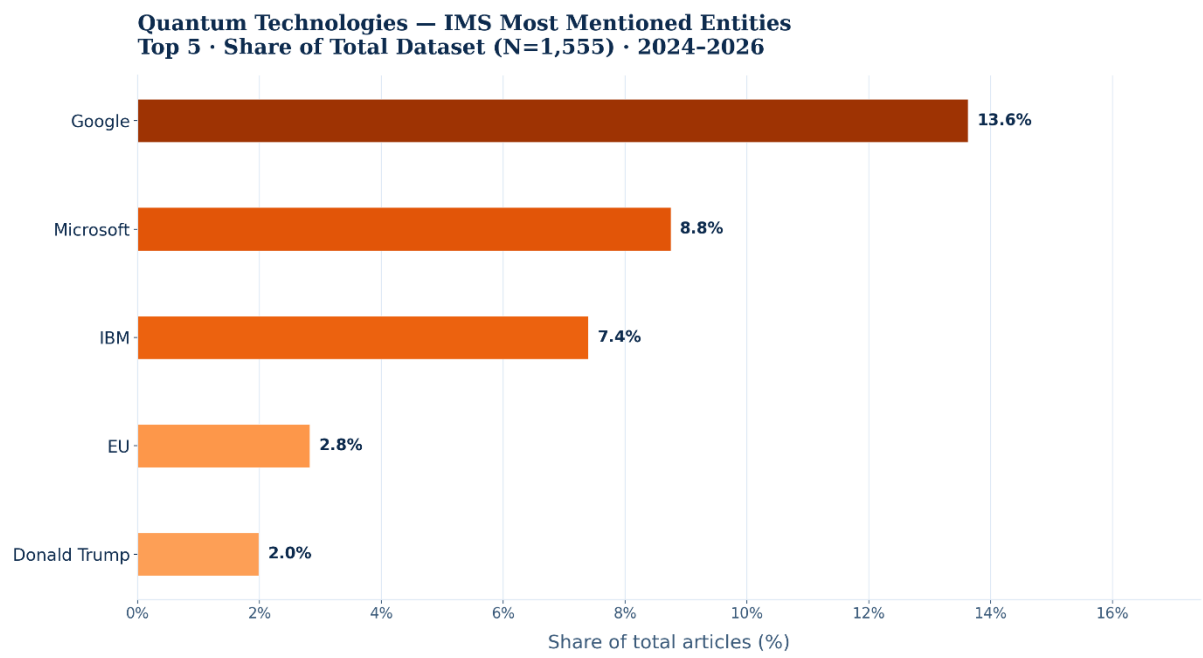
A closer look at entity mentions by article origin reinforces the EU/non-EU contrast already visible in the framing analysis. In EU-origin articles, the EU and the European Commission both rank among the most frequently cited entities (8.6% and 3.6% respectively), keeping European institutional actors visible alongside Google, IBM and Microsoft. In non-EU-origins articles, by contrast, no EU institutional entity appears among the most frequently mentioned; instead, coverage revolve more heavily around market and financial actors, such as Nasdaq Composite, Wall Street, and Motley Fool, as well as state-linked actor such as Xinhua News Agency. This confirms that EU mainstream media sustain a policy-institutional framing of quantum technology even at the entity level, while non-EU mainstream media anchor their coverage more firmly in market and corporate actors. A parallel pattern emerges when looking at the geographical origin of the entities mentioned, independent of where the articles itself was published. In EU-origin articles, 56.0% of entity mentions relate to organisation or individuals based in EU countries, compared with only 22.4% in non-EU origins articles. Conversely, China-linked entities account for 36.0% of mentions in non-EU origin coverage, but only 8.8% in EU-origin coverage. Mentions of US-based entities remain comparably high in both groups (28.0% in EU-origin articles versus 29.1% in non-EU origin articles), confirming that major US technology companies function as a shared reference point across all mainstream media, regardless of the outlet's country of origin.

Key entities in Information Manipulation Set

Figure 5 shows the top 5 most frequently mentioned entities in media coverage by IMS sources. Similar to mainstream media, the most frequently cited entities are the technology firms **Google** (13.6 %), **Microsoft** (8.5 %) and **IBM** (7.4 %). The **EU** appears with a share of 2.8 %, placing it fourth among top 5 entities, yet, rather than leading, its presence in quantum-technology narrative remains behind a landscape dominated by US tech companies.

The US President **Donald Trump** is mentioned in 2 % of the items; he is portrayed as a catalyst for high-profile US-UK agreements on artificial intelligence, quantum computing and nuclear energy, while also being presented as a polarising figure linked to talent outflows, diplomatic frictions and shifting policy commitments.

Figure 5. Share of most cited entities of total IMS coverage on quantum technologies



Source: 1,555 total Quantum Technologies IMS sources · December 2024 - April 2026
Source: IMS. Period: 01-12-2024 to 27-04-2026.

Comparing the mentioned entity across source countries reveals a similar EU/non-EU contrast in the IMS dataset. In EU-origin IMS articles, alongside the leading technology firms (Google 17.1%, IBM 10.3%, Microsoft 10.3%), the EU and European Commission, both feature prominently (6.2% and 2.1% respectively), together with research- and science-oriented entities such as IQM Finland, Nature Communications, the Nobel Foundation and CNRS, pointing to a narrative anchored in European research institutions and quantum-industry actors.

In non-EU-origin IMS articles, the EU still appears but at a lower share (4.0%), while entities linked to geopolitics and security rise to prominence: Donald Trump (3.5%), NATO (1.6%), the Pentagon (1.5%) and Xi Jinping (1.4%), alongside Chinese state-linked entities such as Xinhua News Agency and the Chinese Academy of Sciences. This entity-level pattern reinforces the framing analysis above, where Security and Defence emerged as a distinctly more prominent frame in non-EU IMS coverage (11.7%) than in EU mainstream reporting (4.1%): non-EU IMS sources tend to cast quantum technology within a geopolitical and strategic-rivalry narrative, while EU-origin IMS coverage remains more oriented toward research, industry and institutional policy actors.

Emerging Quantum Technologies: Insights from Innovation Data

This chapter provides an overview of frontier developments in quantum technologies based on innovation data covering the period from 1996 to 2025¹² scientific publications from the Scopus database (Elsevier)¹³, worldwide patents filings from Patstat (EPO)¹⁴ and EU-funded R&D projects from Cordis.¹⁵ Datasets covering emerging technologies in the field have been compiled following the methodology developed by the JRC.T.5,¹⁶ which relies on advanced text mining techniques and biblio-/techno-metric indicators. The quantitative horizon scanning for the current study resulted in a list of 193 ‘weak signals’ of emerging technologies, which have been subsequently grouped in broader technology clusters and assigned to quantum domains.

The analysis in this chapter covers emerging technologies and research developments that gained traction within the innovation community in the most recent years and includes an assessment of their development stage as well as a benchmark of the relative position of major global players in the field. The analysis therefore complements the perspective on quantum reported in the media and illustrate the state of research and innovation in quantum technologies.

1. Emerging research directions and technologies

Table 1 provides an overview of the distribution of detected emerging technologies by broader research areas, grouped by domains Communication, Computing, and Sensing.

Table 1. Weak signal distribution by quantum domain and technology area

Quantum Domain	Technology Cluster	# Emerging Technologies
Communication	Quantum Networks & QKD	19
	Post-Quantum Cryptography & Secure Comms	27
Computing	Quantum Annealing & Optimisation	6
	Quantum Error Correction	14
	Quantum Hardware and Computing Platforms	24
	Quantum Machine Learning & AI	30
	Quantum Software & Development Tools	10
	Quantum Thermodynamics & Energy Devices	2
	Quantum Workforce & Ecosystem	3
	Topological Quantum Materials & Devices	13
	Variational & Hybrid Quantum Algorithms	12
Sensing	Quantum Sensing, Imaging & Metrology	29
	Quantum Materials & Photonic Sources	4

Note: 193 weak signals were identified in the analysis of innovation data. These were clustered based on semantic co-appearance in datasets, applying the Leiden Community detection algorithm. Initial clusters were further grouped in higher level clusters using Antrophic's Sonnet4.5 accounting for broader research areas and communities. The final technology

¹² Scopus Spring 2026 release, PATSTAT Autumn 2025 release, Cordis Spring 2026 release. Data for the most recent years might be incomplete due to indexing lags. Additionally, for patent data an average 18-month lag due to delays between patent application and publication by patent offices must be taken into account.

¹³ Spring 2026 data release – <https://www.elsevier.com/products/scopus>

¹⁴ <https://www.epo.org/en/searching-for-patents/business/patstat>

¹⁵ <https://cordis.europa.eu/>

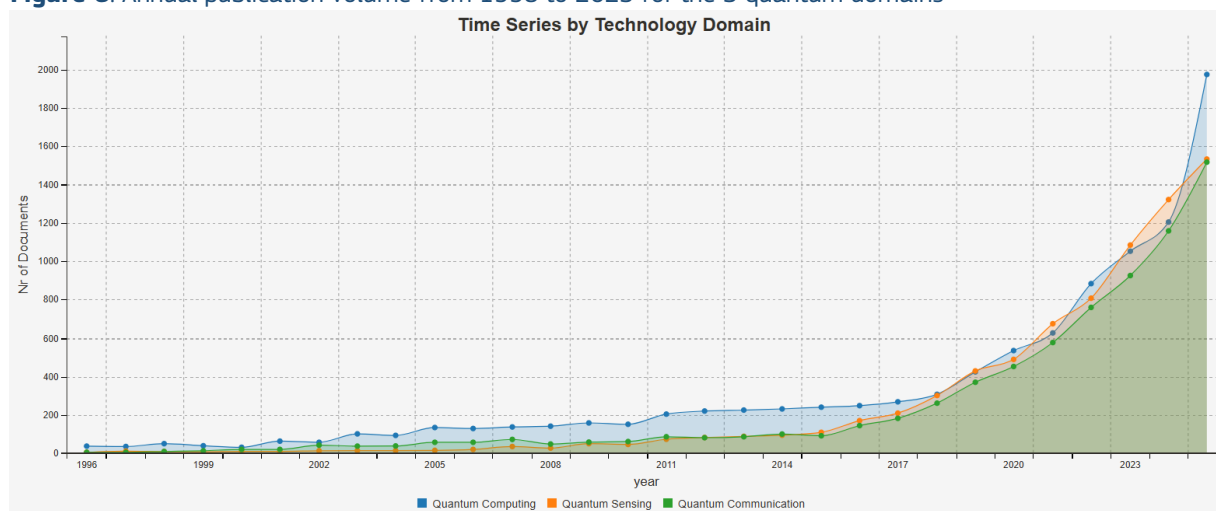
¹⁶ See Methodology section in the Annex for more details.

clusters were classified into broad quantum domains – Communication, Computing, Sensing – by the authors supported by LLMs.

As illustrated in Figure 6, all three Quantum domains experience strong growth over the last 30 years, especially starting from years 2016-2017, with a sharp increase in output volume likely due to the combination of system level factors such as the policy-driven support and infrastructure that embed the NISQ¹⁷ approach into the research ecosystem, cloud-accessible quantum hardware, national quantum initiative launches, and sustained investment inflows. The three domains, however, differ in terms of scale, maturity and recent growth dynamics according to the latest available data¹⁸.

Quantum Communication and **Quantum Sensing** display the strongest growth over the observation period, both with about 22% Compound Annual Growth Rate (CAGR). Both start from a nearly zero document base in the early years, grow substantially over the period 2013 to 2021 (~30% CAGR 2018-2021) and decelerate in the most recent years (about -4pp Communication, -7pp Sensing). **Quantum Computing** is the domain with the longest history and the most established (10,001 documents compared to 7754 Sensing and 7334 Communication). Starting from a relatively higher initial base, it grows with a sustained 15% rate over the full period. Compared to Sensing and Communication, Computing grows modestly in the earliest period, even decelerating slightly post 2013, before accelerating again sharply in the most recent years (30% CAGR post 2021). Computing's recent growth, however, is concentrated in quantum machine learning and AI, variational and hybrid algorithms and software rather than hardware platforms, indicating the effort of extracting near term value from current NISQ devices.

Figure 6. Annual publication volume from 1996 to 2025 for the 3 quantum domains



Source: JRC analysis.

¹⁷ Noise Intermediate Scale Quantum (NISQ) denotes the present-day quantum-computing era characterised by processors that possess tens-to-a-few-hundred qubits but lack full error correction, so they are susceptible to noise and decoherence while still enabling the exploration of quantum advantage for specific tasks. The term was coined by the physicist John Preskil in a highly influential paper in 2018 DOI: [10.22331/q-2018-08-06-79](https://doi.org/10.22331/q-2018-08-06-79).

¹⁸ Data for the most recent years should be considered as provisional as still subject to indexing lags.

Table 2. Document distribution across four-time windows, by quantum-technology domain (1996–2025)

Domain	Total (1996–2025)	Share 1996–2012	Share 2013–2017	Share 2018–2021	Share 2022–2025
Quantum Communication	7,334	9.7%	8.2%	22.6%	59.5%
Quantum Sensing	7,754	5.7%	8.6%	24.4%	61.2%
Quantum Computing	10,001	17.8%	12.1%	18.9%	51.2%

Source: JRC analysis.

Table 3. Compound annual growth rate for communication, computing, and sensing from 1996–2025

Domain	CAGR 1996–2025	CAGR 1996– 2012	CAGR 2013– 2017	CAGR 2018– 2021	CAGR 2022– 2025	Δ CAGR w1→w2 (pp)	Δ CAGR w2→w3 (pp)	Δ CAGR w3→w4 (pp)
Quantum Communication	21.8%	19.0%	20.6%	30.3%	25.9%	+1.6	+9.7	-4.4
Quantum Sensing	21.8%	19.1%	24.1%	30.9%	23.8%	+5.0	+6.8	-7.1
Quantum Computing	14.7%	11.8%	4.5%	26.9%	30.7%	-7.3	+22.4	+3.8

Source: JRC analysis.

2. Technology maturity

The three quantum technology domains present a differentiated profile in terms of technology maturity and research directions. As table 4 shows, the field of quantum technologies in general is concentrated at the “proof-of-concept to validated prototype” stage (48% of emerging technology signals at Technology Readiness Level (TRL¹⁹) 3–5), indicating a landscape that has moved decisively beyond pure theory but has not yet achieved the scale and reliability required for broad deployment.

Table 4. TRL distribution of detected emerging technology signals

TRL	Description	# Em. Technologies	Share
TRL 1–2	Conceptual/Fundamental	14	7.3%
TRL 3–5	Proof-of-Concept to Validated Prototype	93	47.7%
TRL 6–7	System Demonstration	62	32.1%
TRL 8–9	Operational/Deployment	25	13.0%

Source: JRC analysis.

Estimated TRLs, however, are unevenly distributed across quantum domains (cf. **Figure 8**).

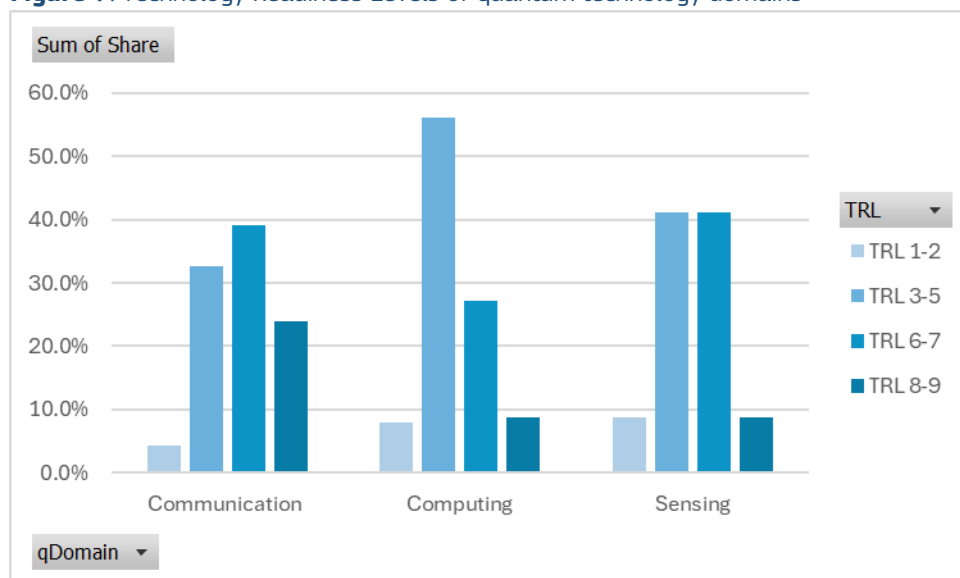
- Frontier research in **Quantum Sensing** equally splits between mature technologies, with some applications, such as atomic clocks, fully deployed and others in advanced stages of

¹⁹ Technology Readiness Levels (TRLs) indicate technology maturity, ranging from TRL 1 (idea) to TRL 9 (full deployment). They are commonly grouped as: TRL 1–2 refers to concept/research; TRL 3–5 means development/ validation; TRL 6–7 describes demonstration/optimisation; TRL 8–9 indicates deployment/implementation. (for more details, see the methodology in the annex).

development, while simultaneously addressing current bottlenecks (e.g. quantum gyroscope) and pursuing more speculative topics (e.g. quantum radar)

- **Quantum communication** also appears as close to real-world application, with **the Post-Quantum Cryptography** cluster **close to deployment** (70% of signals at the system demonstration stage (TRL 6-7) or above and 0 signals at the laboratory stage).
- **Quantum Computing** shows a structural divide between near-term value extraction from current imperfect hardware and the long-term goal of fault-tolerant quantum computers. Fault-tolerant devices remain at the “proof of concept to prototype stage” (TRL 3-5), reflecting technical constraints in current capabilities, while machine learning, hybrid algorithm, and optimization are rapidly progressing towards system demonstration and deployment.

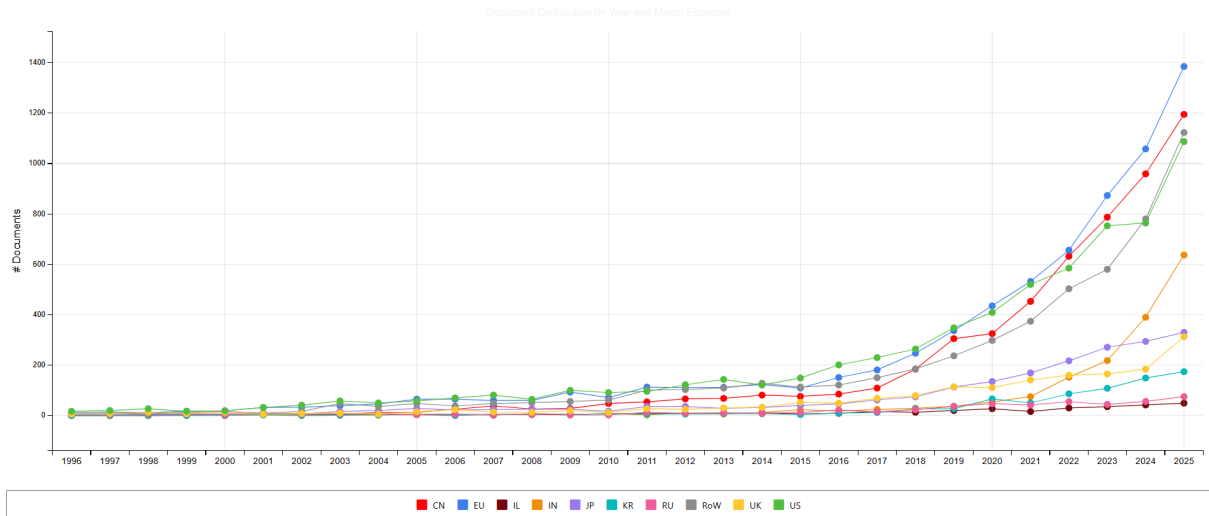
Figure 7. Technology Readiness Levels of quantum technology domains



Source: JRC analysis.

3. Geographic distribution

Figure 8. Document distribution by year and macro economy



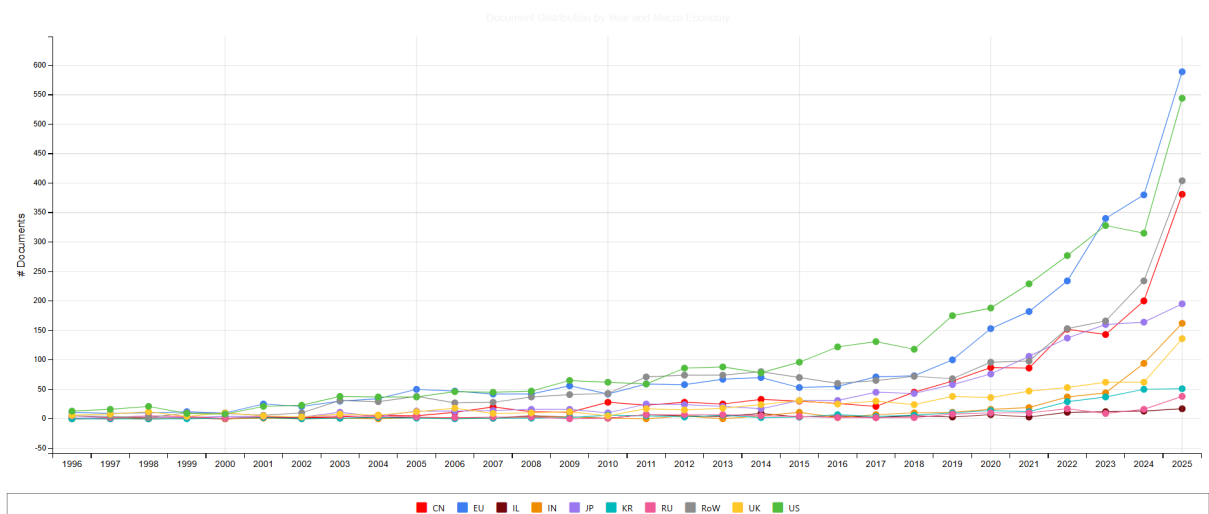
Source: JRC analysis.

In terms of growth dynamics, the EU, China and the United States display similar grow rates, with a marked acceleration over the past 10 years. Other players, including the United Kingdom, Japan, South Korea and India, follow a similar growth trend, although they accelerate more recently (from ~2020) compared to the three major players. The sharp increase in total output growth from a nearly zero base for India is notable and indicates the country's decisive entry in the quantum technology field.

The three quantum technology domains display distinctive profiles in terms of geography, including major R&D hubs, key main players and collaboration networks.

Quantum computing

Figure 9. document distribution by year and macro economy in quantum computing



Source: JRC analysis.

The US shows clear leadership in quantum-computing, leading the top rankings in both science and innovation. The US produces the most impactful research, and hosts the most influential research organisations in the global collaboration network (Figure 11). Its scientific prominence translates in the strongest innovation output, with US organisations dominating overall and in delivering high value patent filings (IP5)²⁰. US companies such as Google, Microsoft, IBM, Dell, Northrop Grumman, are well represented in the list of top patenting organisations to, reflecting a well-balanced innovation ecosystem in which academia, industry and government work closely together.

The **EU** demonstrates a strong and high-impact research presence, ranking second in overall publication volume and rivalling with the US in several sub-fields, including quantum annealing and optimisation, quantum machine learning and AI, and quantum hardware. European organisations are well embedded in the high impact research networks, with two European universities, TU Delft (NL) and Innsbruck University (AT) appearing in the top10 list of most impactful research organisations, otherwise dominated by US institutions. However, the near absence of European industrial partners is noticeable compared to the US innovation ecosystem. Consequently the EU ecosystem relies heavily on US hardware platform providers which serve as the principal experimental infrastructure for advancing research and innovation in the field

China, by contrast, displays a marked mismatch between quantity and impact: while it publishes a large number of papers and files many patents, these outputs achieve relatively low citation impact and delivered few high-value patents (**Figure 11**). The Chinese research network is largely domestic and peripheral to the world-leading research community, with topological-materials work as the only notable exception. The specialisation index²¹ indicates China's focus on near-term value extraction and domestic ecosystem building rather than fundamental research towards fault tolerant quantum computing, with specialisation in Quantum Machine Learning and AI, Variational and Hybrid Quantum Algorithms, and Quantum Software.

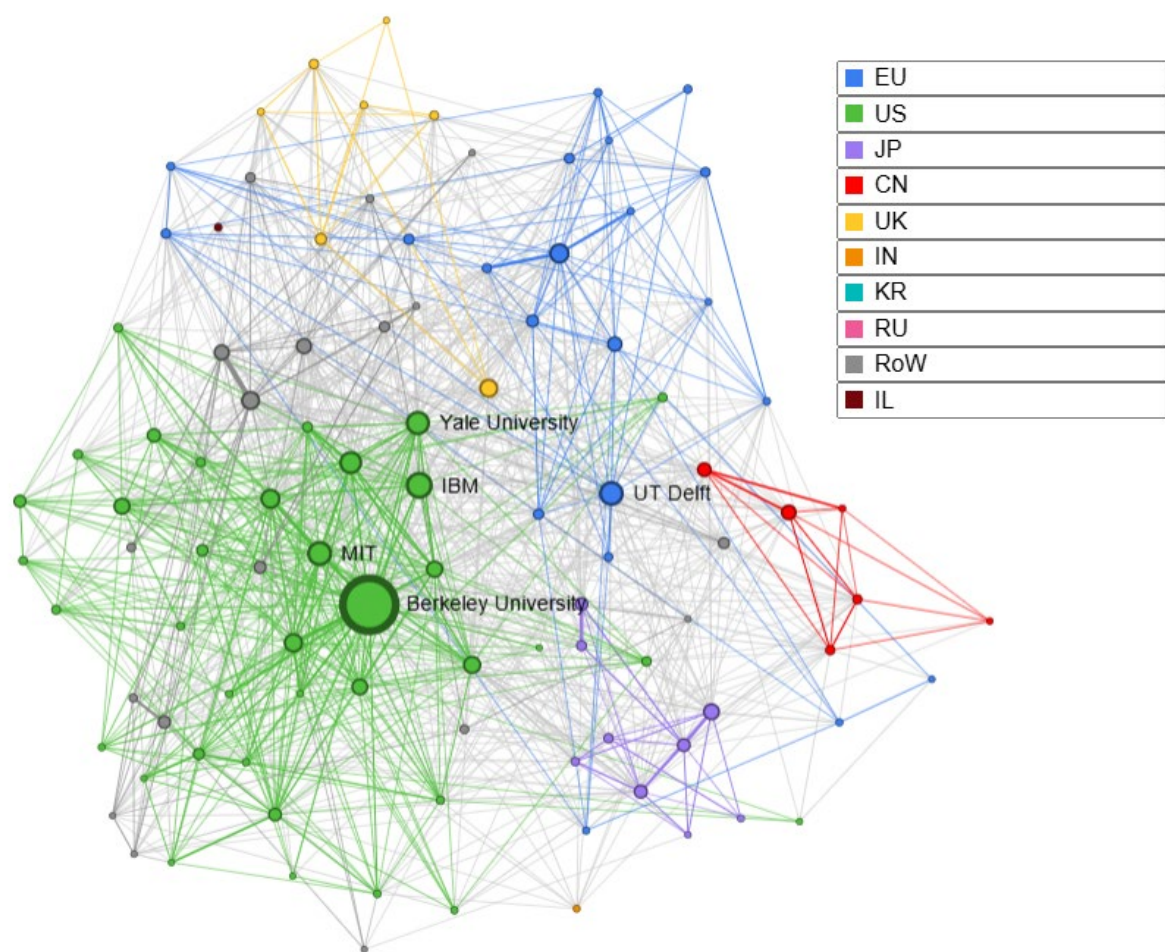
Smaller but high-impact contributors include the **UK, Israel and Japan**, which overall concentrate on fault-tolerant areas (error correction, hardware, topological materials). **Japan's** national innovation ecosystem is the second largest community in the Annealing top cited network, relying on strong industry engagement and national corporations participating alongside leading academic institutions. **Israel** intensely collaborates with the US as its main R&D partner, but maintains ties also with the EU, India and China.

India leverages its strong ICT sector to secure the second-largest publication volume in quantum-machine-learning/AI and the third-largest share of highly cited research, together with notable activity in software and quantum thermodynamics. Among the remaining countries (RoW), **Canada** deserves special mention, with relevant presence across annealing (D-Wave Systems ranks second in the list of top cited research organisations), quantum error correction, quantum hardware, quantum ML&AI, and variational algorithms. **Switzerland** likewise demonstrates a notable research footprint, with its organisations appearing consistently among the highest-ranked institutions across several quantum computing fields.

²⁰ IP5 patents are defined as patent families filed in at least two patent offices worldwide, of which one of the five largest intellectual property offices: the European Patent Office (EPO), the Japan Patent Office (JPO), the Korea's Ministry of Intellectual Property (MOIP), the National Intellectual Property Administration of the People's Republic of China (CNIPA), and the United States Patent and Trademark Office (USPTO).

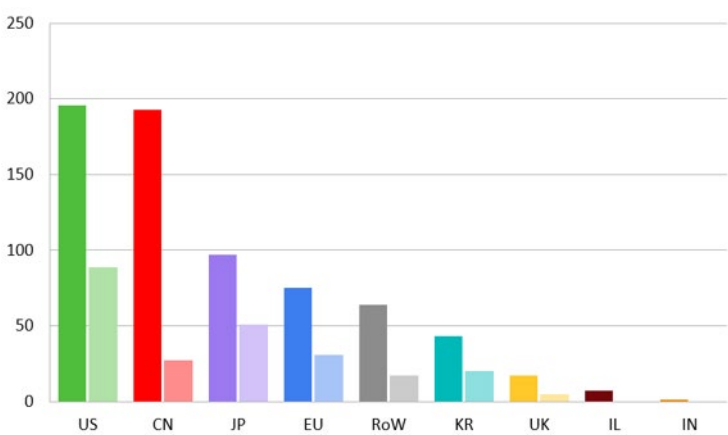
²¹ Revealed Technology Advantage (RTA) index.

Figure 10. Collaboration network in high impact research in quantum computing



Source: JRC analysis.

Figure 11. Patent (dark) and High Value (IPS) (light) patents by Macro Economy



Source: JRC analysis.

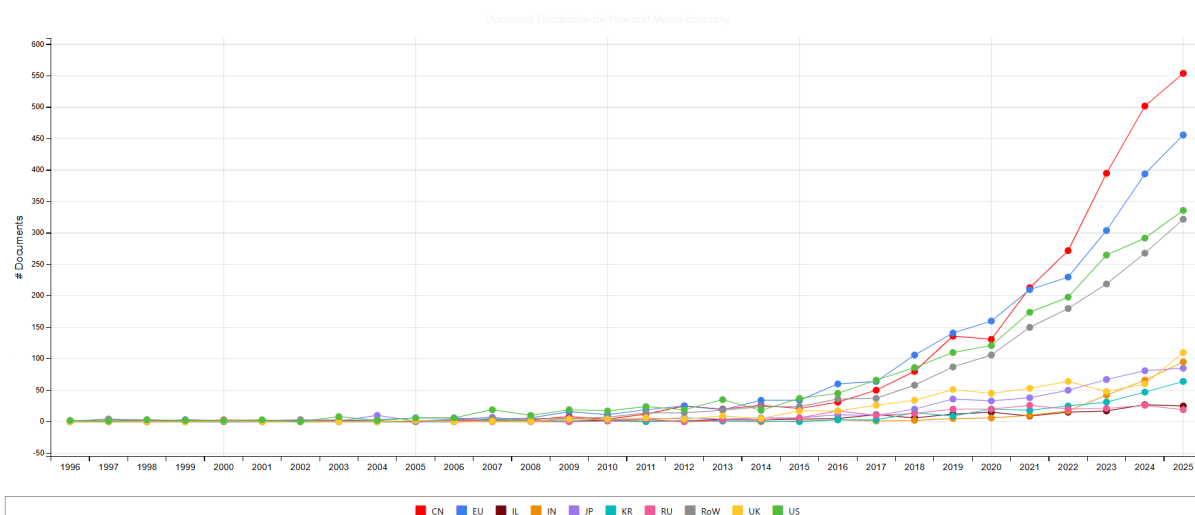
Table 5. Top20 patenting organisations

Organisation	# Patents	High Value	% High Value	Country	Macro Economy
NTT Corporation	31	17	54.8	Japan	JP
Google	26	13	50	US	US
D Wave Systems	19	6	31.6	Canada	RoW
Riverlane	19	3	15.8	UK	UK
Microsoft	13	11	84.6	US	US
NEC Corporation	13	6	46.2	Japan	JP
Fujitsu	11	8	72.7	Japan	JP
Xidian University	11	0	0	China	CN
Dell Products	11	0	0	US	US
IBM	11	6	54.5	US	US
Yale University	10	8	80	US	US
Korea Electronics Telecomm	10	3	30	South Korea	KR
Toshiba Corporation	9	4	44.4	Japan	JP
Waseda University	9	4	44.4	Japan	JP
Huawei	9	7	77.8	China	CN
Tencent	9	9	100	China	CN
UT Qingdao	9	0	0	China	CN
MIT	8	5	62.5	US	US
Alice and Bob	8	2	25	France	EU
Harbin Institute of Technology	8	0	0	China	CN

Source: JRC analysis.

Quantum Sensing

Figure 12. Document Distribution by Year and Macro Economy (Quantum Sensing)



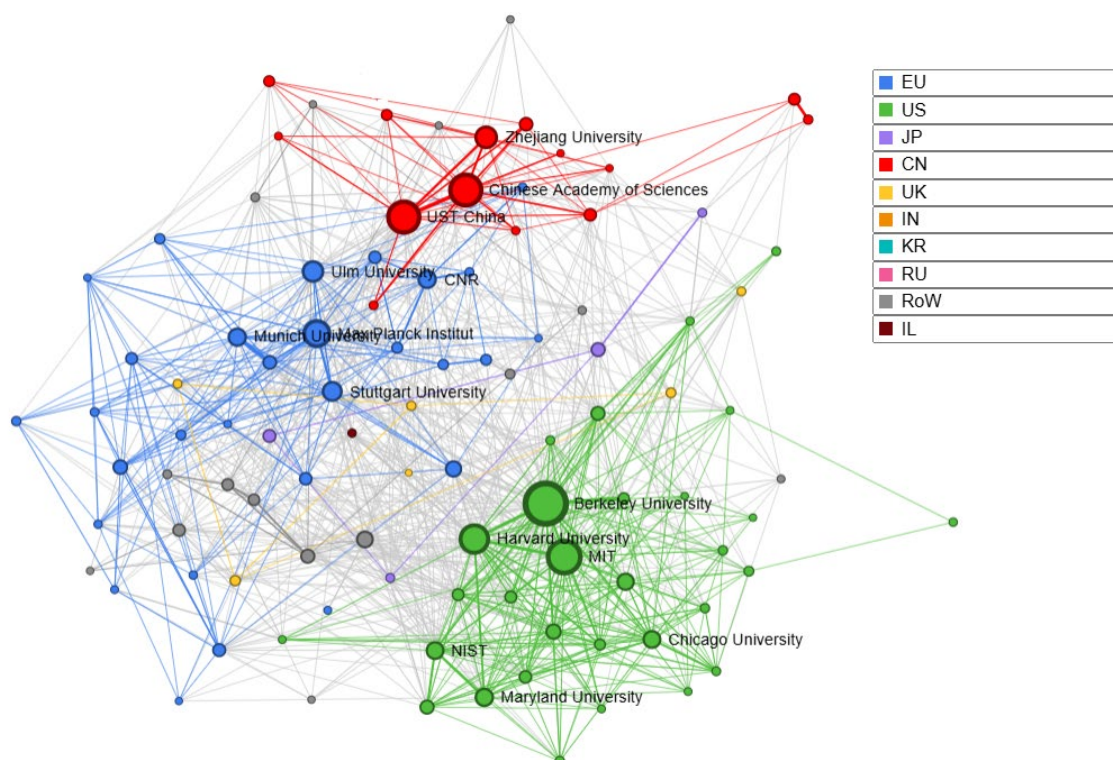
Source: JRC analysis.

The **EU** leads on overall scientific publication volume (2,167), with the **US** close behind (1,855) but ahead on impact quality. Similarly to Computing, **China** shows a volume-impact gap: second by publication and patenting volume and with the sharpest increase in research output starting from 2020, China exhibits markedly lower international impact compared to the EU and US. US and China alternatively lead the ranking of top research and patenting organisations in terms of impact and volume respectively. European organisations are well represented in the most influential research

ranking (**Figure 13**), while very few European organisations make it to the top ranking for patent filings.

Among smaller players, the **UK** stands out for both volume and impact in science, ranking fourth overall and exhibiting citation performance comparable to that of US and EU. **Japan, India** and **Israel** also appear as notable scientific contributors. In the rest of the world group, **Australia** and **Switzerland** appear as prominent hubs.

Figure 13. Organisation Network in High Impact Research in Quantum Sensing



Source: JRC analysis.

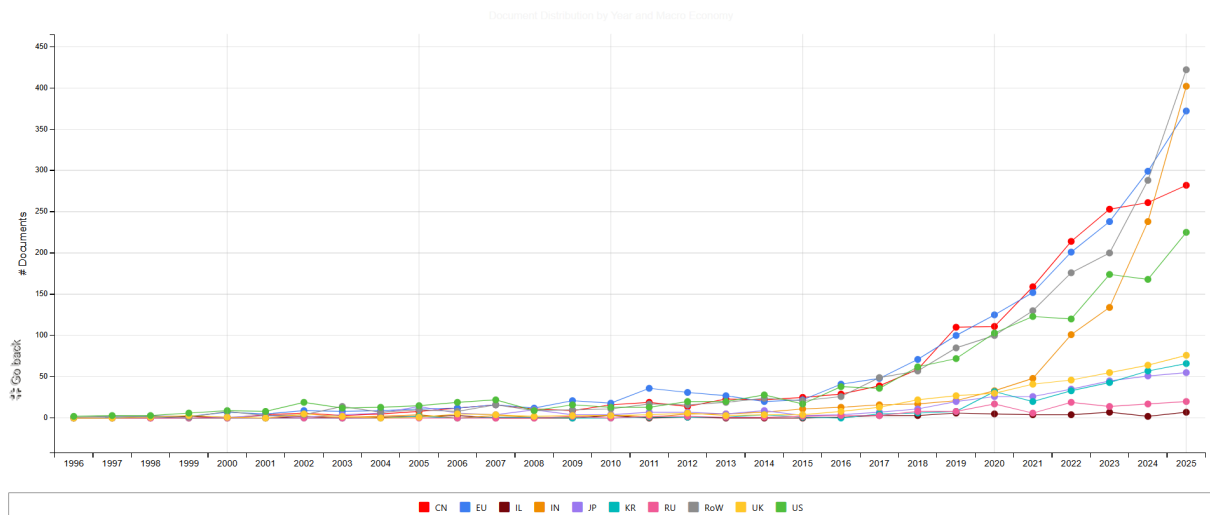
Table 6. Top20 patenting organisation in quantum sensing emerging technologies

Organisation	# Patents	IP5 count	% IP5 share	Country	Macro Economy
Chicago University	5	4	80.0%	United States	US
Sca Hygiene Prod	7	3	42.9%	Sweden	EU
MIT	6	3	50.0%	United States	US
Anhui Guosheng Quantum Tech Co	24	2	8.3%	China	CN
Agency for Defense Development	7	2	28.6%	South Korea	KR
Toshiba Corporation	4	2	50.0%	Japan	JP
Hong Kong University	2	2	100.0%	China	CN
Halliburton Energy Services Inc	2	2	100.0%	United States	US
Pusan National University	2	2	100.0%	South Korea	KR
Maxxon Corp	2	2	100.0%	United States	US
Dell Seven Inc	2	2	100.0%	United States	US
Southeast Nonwovens Inc	2	2	100.0%	United States	US
Kimberly Clark Co	2	2	100.0%	United States	US
Zhejiang University	9	1	11.1%	China	CN
UST China	9	1	11.1%	China	CN
State Grid Corp China	8	1	12.5%	China	CN
Ulsan National Institute of Science and Technology	6	1	16.7%	South Korea	KR
Colorado University	3	1	33.3%	United States	US
Japan Science and Technology Agency	3	1	33.3%	Japan	JP
Nottingham University	3	1	33.3%	United Kingdom	UK

Source: JRC analysis.

Quantum Communication

Figure 14. Document distribution by year and macro economy in Quantum Communication



Source: JRC analysis.

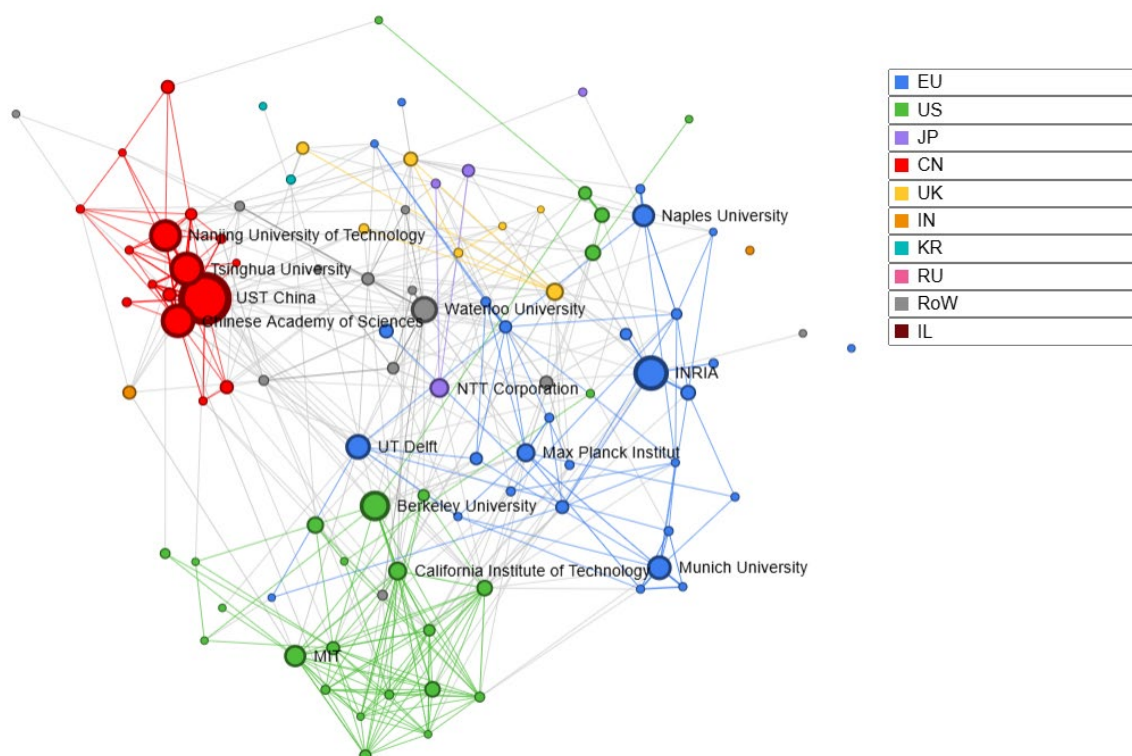
The **EU** leads in scientific output in both volume and impact, with European organisations occupying central positions in the high impact research network and strong European representation in the top ranking of most influential organisations. Similarly to Computing and Sensing, however, Europe lags behind other economies when it comes to translate scientific prominence to commercial value, with few patents filed by European organisations.

Despite ranking only fourth in scientific publication volume, the **US** ranks second in high-impact research and first in high-value patent filings, suggesting that its innovation ecosystem focuses on high-level influential research and high-value inventions rather than mere quantity.

The gap between volume and impact for **China** is generally confirmed in the Communication domain, both for scientific and innovation output. Nevertheless, four Chinese institutions top the ranking of high impact research organisations, suggesting a strong concentration of research excellence in a limited number of flagship institutions.

India and the **UK** also make significant contributions to scientific output in both volume and impact. NTT Corporation features among the most highly cited research organisations, underscoring the strong engagement of **Japan's** industrial actors in quantum technology research.

Figure 15. Organisation Network in High Impact Research in Quantum Communication



Source: JRC analysis.

Table 7. Top20 patenting organisations in Quantum Communication Emerging Technologies

Organisation	# Patents	IP5 count	IP5 share (%)	Country	Macro economy
PQSecure Technologies	7	7	100.0%	United States	US
ISARA	5	3	60.0%	Mexico	RoW
Microsoft	5	2	40.0%	United States	US
Sun Microsystems Inc	2	2	100.0%	United States	US
Quantum Base Ltd	2	2	100.0%	United Kingdom	UK
Anametric Inc	2	2	100.0%	United States	US
Pcgee Pte Ltd	2	2	100.0%	Singapore	RoW
Nanjing University of Posts and Telecommunications	6	1	16.7%	China	CN
STMicroelectronics	3	1	33.3%	Italy	EU
State Grid Fujian Electric Power Co Ltd	2	1	50.0%	China	CN
Korea Electronics Telecomm	2	1	50.0%	South Korea	KR
Siemens	2	1	50.0%	Germany	EU
Hainan University	2	1	50.0%	China	CN
IBM	2	1	50.0%	United States	US
Qusecure	2	1	50.0%	United States	US
KAIST	2	1	50.0%	South Korea	KR
MIT	2	1	50.0%	United States	US
British Telecommunications	2	1	50.0%	United Kingdom	UK
Hewlett Packard	1	1	100.0%	United States	US
Shanghai Zhaoxin Semiconductor Co Ltd	1	1	100.0%	China	CN

Source: JRC analysis.

Comparative Analysis of Media narratives and Scientific Landscape

The analysis reveals a consistent gap between media coverage and innovation activity. Media coverage primarily concentrates on high-profile hardware announcements – such as the unveiling of new quantum chips by major corporations like Google and Microsoft – emphasising their potential to revolutionise industries and conveying a perception of imminent transformative change. This attention aligns overwhelmingly with the quantum computing domain, and within it with fault-tolerant hardware that remains far from deployment. Innovation data, by contrast, show quantum communication and quantum sensing to be comparably active to quantum computing and, on several indicators, closer to application. The pattern reflects the selection logic of news rather than the distribution of technical progress: corporate breakthrough claims arrive as discrete, dateable events, whereas standardisation and gradual deployment generate few comparable news moments and therefore less coverage.

In the landscape of post-quantum cryptography, media narratives often describe an urgent need to address potential threats that quantum computing could pose to conventional cryptographic systems. The urgency is grounded in the projected capabilities of quantum computers to break existing encryption methods, posing significant risks to cybersecurity.

However, scientific evidence indicates that the hardware required to realise such machines remains far from current capabilities, with fault-tolerant quantum computers still at TRL 3-5 and several competing hardware approaches currently being explored, none of which has yet demonstrated the scale or error rates required. Moreover, solutions such as lattice-based cryptographic algorithms are already in the deployment phase. The urgency conveyed by media coverage is therefore real but mislocated: while the threat of capable quantum computers is genuine but not imminent, the transition to post-quantum cryptography remains time-critical because full migration will require several years of sustained effort to complete.

Quantum sensing, identified as the most mature domain in quantum technologies, receives comparatively less media coverage. Despite its readiness for practical deployment and applications in fields like navigation and metrology, sensing rarely appears in reporting, which favours experimental developments in quantum computing. Developments in the sensing domain, such as quantum accelerometers, quantum gravimeters, inertial navigation systems and cold atom quantum sensors, are closer to real-world deployment, and worth monitoring given their relevance for future GNSS-independent navigation systems, Earth observation as well as their potential dual use applications.

When it comes to the performance of the major economic players, media coverage tends to emphasise achievements by US corporations while overlooking the diversity and complexity of the global quantum innovation ecosystem. A comparison with innovation data reveals a more balanced picture, in which leadership varies across technology domains and cannot be reduced to a single dominant actor. This is particularly true for quantum computing, where the field remains in a fluid phase of technological development. Multiple hardware platforms and computing paradigms are being pursued in parallel, and no single technological approach has yet emerged as the dominant pathway towards scalable, fault-tolerant quantum computing.

Sources within the Information Manipulation Set (IMS) exploit the technical complexity of quantum technologies to amplify geopolitical threat perceptions, such as quantum computing as a tool for coercion, while downplaying European advancements and stressing US decline regarding quantum technologies. By contrast, the innovation data does not support these narratives. The US remains the leading actor overall, particularly in quantum computing, supported by a strong industrial ecosystem. At the same time, the EU rivals the US in scientific excellence and even outperforms it in specific

domains, such as quantum annealing and optimisation, although it lacks comparable industrial anchoring. China's position, while significant, appears more limited and domain-specific than portrayed by IMS narratives, whereas Russia does not emerge as a major player across the scientific and innovation indicators considered. This further undermines the public's ability to accurately assess the state of quantum technologies, highlighting the need for more nuanced media coverage or complementary information dissemination that reflects the genuine state of the scientific landscape.

Conclusions

The report presents the results of an analysis of news coverage on quantum technologies, from mainstream and Information Manipulation Sources in the EU, the US, China and Russia. This is complemented by an assessment of the underlying scientific progress and the key actors driving the field from 1996 until 2025.

Based on the analysis of **media coverage** from 1 December 2024 to 27 April 2026, this report concludes the following regarding the topic of quantum technologies:

- **Peak reporting** – The analysis shows a marked contrast between mainstream media and the information-manipulation set (IMS) in their reporting on quantum technologies. Mainstream outlets primarily foreground announcements from large corporations such as Google and IBM, reflecting the commercial milestones that dominate the public discourse. In contrast, IMS sources mainly highlighted the active role of Russia and China by stressing solutions and milestones achieved by Russian and Chinese scientists.
- **Thematic focus** – Mainstream coverage is centred on technological breakthroughs and applications, research collaborations, investments, commercialisation and market growth, European and global strategies/policies and associated challenges. IMS narratives highlighted perceived Chinese dominance, US decline, European marginality, quantum computing threats to global stability, European reliance on US/Chinese technology and its use for geopolitical coercion.
- **Sentiment** – Positive sentiment prevailed in both mainstream and IMS sources, suggesting that media reporting favours the promise and market potential of quantum technologies over their limitations and risks.
- **Framing dimensions** – Both media groups primarily framed stories around *Capacity and Resources* and *Policy Prescription and Evaluation*, emphasising infrastructure, funding and human capital required for Europe's quantum agenda, together with regulatory measures. Security concerns were more prominent in non-EU IMS sources.
- **Key actors** – The most frequently mentioned entities were the technology firms Google, IBM and Microsoft, which drive quantum computing development. The EU appears as the fourth-most cited actor, mainly portrayed in the context of its ambition to become a global quantum leader.

The analysis of **innovation data** shows a robust and expanding scientific output across the three quantum domains: communication, computing and sensing. The US remains the most influential player in quantum computing overall, contributing the largest share of highly cited research. The EU challenges this leadership in specific areas of quantum computing, such as annealing and optimization, while leading in quantum sensing. China's rapid growth in innovation output has not yet translated into comparative international impact, except in quantum communication where Chinese institutions figure among the most influential organisations globally.

Media coverage, by contrast, is heavily skewed toward high-profile hardware announcements and commercial milestones, particularly from US companies such as Google and Microsoft. The urgency media outlets convey about the threat posed by future fault-tolerant quantum computers to national security is real but mislocated: quantum computing hardware remains at the proof-of-concept to laboratory-validation stages, with fault-tolerant machines far from deployment, while the genuinely

time-critical task - the migration to post-quantum cryptography, given 'harvest now, decrypt later' strategies - receives little attention. Quantum sensing, the most mature domain, with technologies progressing from prototype demonstration to operational deployment in navigation, metrology, and Earth observation, likewise receives comparatively little coverage. The overall picture available from media coverage alone thus differs substantially from the one described by the innovation data, including regarding Europe's position in the global innovation landscape.

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Annexes

Annex I – Methodology for media analysis

This science media intelligence briefing is part of the series of joint publications of the European Science-Media Hub (ESMH) and the JRC Text and Data Mining Unit, covering topics relevant for specific policy domains, ranging from water scarcity, genetically modified organisms, and new-genomic techniques to artificial intelligence in science communication as well as in healthcare. For further information please consults <https://sciencemediahub.eu/science-media-intelligence-reports/>.

Data selection

Articles relevant for the analysis are selected by searching for a set of relevant keywords or keyword combinations that appear in the title or lead line of the article. Lead line refers to approximately the first 300 characters of the article, following the title. These keywords are translated into all EU24 languages, Chinese and Russian. Additionally, the English keywords were searched for in the English translations of the articles' title and lead line.

The keyword combinations have been selected carefully to include all important articles but to avoid irrelevant 'noise' in the news collection. Besides all 27 EU Member States, the media from the US, Russia and China were analysed.

Query

Quantum computing, Quantum annealing, Quantum for AI, Quantum simulation, Quantum error correction, Noisy Intermediate Scale Quantum, Qubits, Quantum Advantage, Quantum Supremacy, Quantum Utility, Quantum Communication, Quantum key distribution, Quantum sensing, Quantum metrology, Quantum metrology, quantum technologies, Quantum internet, Post-Quantum Cryptography, Quantum Teleportation, Quantum repeaters, Quantum detectors, Quantum MRI, Quantum gravimeter, Quantum magnetometer, Quantum gyroscopes, NISQ device

Source selection

The manually curated list of online media sources, administrated by Europe Media Monitor (EMM)⁵, includes over 25,000 domains worldwide. Of these, more than 2,000 represent the Information Manipulation Set (IMS). The IMS used in this study represents a collection of online open-source news media outlets identified either through the European External Action Service's assessments, as outlined in its reports on Foreign Information Manipulation and Interference Threats²², or by independent fact-checking organisations as potential spreaders of mis- and disinformation. The set also includes structured and coordinated propaganda networks, such as the Russia-linked Pravda network analysed in the *Portal Kombat* report²³ and the related expanded network examined in reports by the Digital Forensic Research Lab²⁴. It is necessary to note that our findings are contingent on the IMS sources monitored throughout the data collection process. The uneven country coverage of our list of IMS sources, with a prevalence of Russian, US, and Western European sources, is contingent on the fact checkers and independent experts that we relied

²² https://www.eeas.europa.eu/eeas/4th-eeas-annual-report-foreign-information-manipulation-and-interference-threats_en

²³ [PORTAL KOMBAT: A structured and coordinated pro-Russian propaganda network](#)

²⁴ [Amaury Lesplingart and Valentin Châtelet, "Russia's so-called "Pravda" network expands worldwide." Digital Forensic Research Lab \(DFRLab\)](#), [Amaury Lesplingart and Valentin Châtelet, "Russia-linked Pravda network cited on Wikipedia, LLMs, and X." Digital Forensic Research Lab \(DFRLab\)](#).

on to compile the list of IMS sources, given that they are mostly focused on the US and Western Europe.

Automated multilingual clustering and generation of cluster title and cluster summaries

Clusters are computed by a community-based algorithm working with titles and texts of the articles in original language. As a metric to compare content of the articles the semantic similarity was used. Text embeddings together with multilingual clustering of embedded texts were used to group articles of similar content into clusters. Each story (cluster of articles) was labelled with a title, summary and overview of the EU portrayal generated by LLM (GPT-4o).

Sentiment classifier

In our sentiment analysis, we utilise a state-of-the-art sentiment model, namely the XLM-RLnews-5 model, which is specifically designed for document-level sentiment analysis across multiple languages. Based on XLM-RoBERTa-Large, this model has been fine-tuned for sentiment analysis using the Unified Multilingual Sentiment Analysis Benchmark (UMSAB) dataset. The sentiment classes are computed on the English translation of the headlines.

More details on the model development are available in:

Di Nuovo, E., Cartier, E., Bertrand De Longueville, 'Meet XLM-RLnews-8: Not Just Another Sentiment Analysis Model'. In *Natural Language Processing and Information Systems, 28th International Conference on Applications of Natural Language to Information Systems, NLDB 2024, Turin, Italy, June 25–27, 2024, Proceedings* (pp. 1). Springer Science and Business Media Deutschland GmbH, 2024.

Named entity recognition

Names Entity Recognition has been done using the NEROne module which has been developed in the context of the Europe Media Monitoring project. For detailed information, see:

Bandelli, L., Cartier, E., Verile, M. (forthcoming, 2026) Named Entity Recognition and Linking: Description of the New System, JRC Publications

Cartier, E., Bandelli, L., Verile, M. (2025) Named Entity Recognition and Linking: state of the art and comparison with the Europe Media Monitor NER/NEL system, JRC Publications

Cartier, E. and Peetermans, E. (2024), Combining deep learning models and lexical linked data: Some insights from the development of a multilingual news named entity recognition and linking dataset, In 'Proceedings of Deep Learning and Linguistic Linked Data, LREC-COLING 2024 Workshop', URL <https://api.semanticscholar.org/CorpusID:269951184>.

Automated framing detection

Automated detection of framing dimensions constituted an extension of our analysis conducted in this report. This work was based on in-house machine learning classifiers that detect framing dimensions in the articles. The machine-learning algorithm trained on a multilingual corpus.

Framing refers to the perspective under which an issue or a piece of news is presented. We consider the following 14 framing dimensions:

Table 8. *Overview of the framing dimensions' meaning*

Framing dimension	Reference of
Economic	costs, benefits or other financial implications.
Capacity and Resources	availability of physical, human or financial resources, and capacity of current systems.
Morality	religious or ethical implications
Fairness and Equality	balance or distribution of rights, responsibilities and resources
Legality, Constitutionally and Jurisprudence	rights, freedoms, and authority of individuals, corporations and government.
Policy Prescription and Evaluation	discussion of specific policies aimed at addressing problems.
Crime and Punishment	effectiveness and implications of laws and their enforcement
Security and Defence	threats to welfare of the individual, community or nation.
Health and Safety	health care, sanitation, public safety
Quality of Life	threats and opportunities for the individual's wealth, happiness, and well-being.
Cultural Identity	traditions, customs , or values of a social group in relation to a policy issue.
Public Opinion	attitudes and opinions of the general public, including polling and demographics.
Political	considerations related to politics and politicians, including lobbying, elections, and attempts to sway voters.
External Regulation and Reputation	international reputation or foreign policy

Source: JRC analysis based on Piskorski et al.

For more information see the JRC Technical Report:

Piskorski, J., Stefanovitch, N., Bausier, V. A., Faggiani, N., Linge, J., Kharazi, S., Nakov, P. (2023). *News categorization, framing and persuasion techniques: Annotation guidelines*. European Commission, Ispra, JRC132862. https://knowledge4policy.ec.europa.eu/text-mining/news-categorization-framing-persuasion-techniques-annotation-guidelines_en

Annex II – Methodology for innovation data analysis

Detection of Emerging Technologies (Weak Signals)

The methodology applied in this study follows the approach developed by Unit T.5 of the JRC to detect ‘weak signals’ of promising early-stage technologies²⁵. The analysis is conducted using the TIM Technology²⁶ platform, an advanced monitoring system integrating innovation data, natural language processing (NLP) techniques and Artificial Intelligence.

Two document corpora are used for the detection of weak signals: the Scopus database of scientific publications from Elsevier²⁷ and the PATSTAT database of patents from the European Patent Office²⁸. Weak signals are identified through a text-mining process that generates a large dictionary of multi-word scientific and technological concepts extracted from titles, abstracts and keywords of recent publications and patent documents. Concepts are subsequently normalised, grouped and ranked by relevance using a modified TF-IDF approach. This dictionary forms the basis for automated semantic queries used to construct document collections across the full-time period from 1996 to 2025.

Two complementary detection processes are applied. The systematic” process generates document collections for each concept in the dictionary. Datasets are then ranked using an activeness indicator, which measures the share of documents published in recent years relative to the full time period. In parallel, a targeted process relies on expert-defined semantic searches to retrieve documents related to specific technological topics and to identify emerging concepts with high recent activeness.

Promising raw weak signals are subsequently reconstructed in TIM Technology through refined search queries incorporating synonyms and alternative terminology to maximise document retrieval. The final selection of weak signals is based on activeness indicators for the period 2022–2025: Signals with activeness between 70% and 100% are automatically retained, while those between 50% and 70% showing strong recent growth are also included based on the assessment of analysts. Using this procedure, 193 weak signals were identified.

Weak signals were subsequently grouped into broader technology clusters using the community detection Leiden algorithm²⁹ to uncover the network structure of frontier research in the field. Finally, technology clusters were assigned to quantum domains by analysts supported by large language models (LLMs) (Anthropic’s Claude Sonnet 4.5).

Technology Readiness Level Assessment

Technology Readiness Levels (TRLs) are indicators of the maturity level of particular technology. This measurement system provides a common understanding of technology status and addresses the entire innovation chain. There are nine technology readiness levels; TRL 1 being the lowest and TRL 9 the highest. TRLs have been grouped in four categories, following common practice in foresight activities in the Joint Research Centre. The four TRL categories are defined as follows: TRL 1-2 - Concept and research: Basic principles are observed, and potential applications are hypothesised. This involves fundamental research and concept formulation; TRL 3-5 - Development and validation: Active development stage including proof of concept, experimental validation, and early prototypes,

²⁵ For further details on the methodology see Eulaerts, O.D., Grabowska, M. and Bergamini, M., *Early stage technologies in the field of Energy*, Publications Office of the European Union, Luxembourg, 2024, [doi:10.2760/711_JRC135619](https://doi.org/10.2760/711_JRC135619).

²⁶ https://knowledge4policy.ec.europa.eu/text-mining/TIM_tools_en

²⁷ <https://www.elsevier.com/products/scopus>

²⁸ <https://www.epo.org/en/searching-for-patents/business/patstat>

²⁹ Traag, V. A., Waltman, L., & Van Eck, N. J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Scientific reports*, 9(1), 5233.

tested in controlled settings; TRL 6-7 - Demonstration and optimisation: Further developed technologies demonstrated in relevant environments. Focus is on addressing remaining technical challenges, refinement and performance optimisation; TRL 8-9 - Deployment and implementation: Fully developed technologies tested and validated, now ready to be or already being deployed or implemented in real-world contexts.

TRLs for the present study were estimated using LLMs through four independent iterations run in separate sessions (two using OpenAI's GPT-4o, two using Anthropic's Claude Sonnet 4.5). A TRL assessment was accepted in first instance where at least three of the four independent estimates agreed (approximately 70% of cases). The remaining cases were resolved through an additional iteration, in which the model was asked to review the previous estimates and provide a reasoned justification for the final assessment. The resulting TRL distribution was validated by analysts within the team. Each LLM assessment was based on the titles and abstracts of the 20 most-cited scientific publications in each dataset, with citation counts normalised by field and publication year.

Annex III – Domain-focused technological landscape

Quantum Computing

The signals detected clearly mark the internal divide of quantum computing research into two different paradigms with different hardware requirements, different time horizons, and different publication dynamics. Research is split between extracting near-term value from noisy intermediate-scale quantum (NISQ) devices and pursuing the long-term goal of fault-tolerant quantum computers.

On the one hand, the focus for **extracting near-term value** is on leveraging current NISQ-devices for practical applications in finance, chemistry, and medicine. Technology fields such as Variational & Hybrid Quantum Algorithms, Quantum Machine Learning, quantum annealing and optimisation, and quantum software tools are growing rapidly (CAGR > 80 % for some fields in 2017-2021). They aim to deliver demonstrable advantages in finance, chemistry, biology and medicine with the limited qubit numbers and short circuits currently available.

On the other hand, the research and innovation community maintains the **long-term goal of fault-tolerant architecture**, which stands beyond current capabilities. The dominant effort is in **Quantum Error Correction** and **fault-tolerant Quantum Hardware and Computing Platforms**. Despite three decades of work, growth is modest (CAGR $\approx$ 16 %) and most signals sit at early-to-mid technology readiness levels. No operational deployments exist yet.

Quantum Sensing

Quantum sensing is the most advanced area of quantum technology, with a literature base of more than seven thousand papers covering imaging, metrology, timing, navigation and photonic detection. Diamond-based nitrogen-vacancy sensors have already reached a stage where commercial instruments such as magnetometers and microscopes are available, and rubidium atomic clocks are fully deployed in global navigation satellite systems and telecommunications networks. Other sub-fields—quantum accelerometers, gravimeters and cold-atom interferometers—have progressed to the point of laboratory demonstration and are being investigated for navigation that does not rely on satellite signals, for defence applications and for Earth-observation from space. Single photonic sensing includes quantum lidar and ghost-imaging devices that are at early demonstration level, while quantum radar remains in an experimental phase because of unresolved microwave-photon engineering challenges. The same single-photon infrastructure also underpins high-maturity time-bin quantum key distribution, illustrating a convergence between sensing and secure communications. Novel material approaches such as metal-organic-framework spin-qubits and magnon-based sensors are still at early research stages, and a rapidly expanding strand of work on quantum materials and photonic sources—characterised by a very high rate of patenting—indicates strong commercial potential despite its recent emergence.

Quantum Communication

Quantum communication is the smallest dedicated quantum-technology field, when excluding the hybrid Post-Quantum Cryptography, treated separately in this chapter. The only cluster in this domain covers different aspects of **quantum networks and quantum-key-distribution** (19 signals, 1160 documents). The infrastructure side includes already mature trusted-node repeaters (operating at the highest readiness level) and satellite-based key-distribution and multi-node terrestrial networks at

demonstrator stage, alongside very early-stage concepts such as network non-locality and large-scale quantum networking. The protocol side shows a clear readiness ladder: current commercial schemes rely on physically secured relay points, limiting security, while frontier work focuses on eliminating those trusted nodes. Demonstrator-level approaches include entanglement-based key-distribution, measurement-device-independent techniques that remove the need to trust detectors, and twin-field key-distribution that extends transmission distance and reduces the number of required relays, all aimed at achieving fully device-independent quantum communication.

Post-Quantum Cryptography cluster

Post-Quantum Cryptography and Secure Comms (PQC) (7 signals, 5,216 documents) is structurally unique as the core objective is not to develop quantum technology but to defend against it.

Its purpose is to safeguard current information-technology systems against future fault-tolerant quantum computers that could break classical public-key schemes. The field expanded sharply after the US National Institute of Standards and Technology launched its standardisation competition in 2016³⁰, nearly doubling output in the five years that followed and maintaining a 22 % compound annual growth rate between 2021 and 2025. Standardised lattice-based algorithms such as Kyber, Dilithium and FALCON have reached the highest technology-readiness levels (8-9) and are already being deployed, while alternative code-based and isogeny-based schemes remain at intermediate readiness (6-7). The research frontier (TRL 3-5) is occupied by post-quantum homomorphic encryption, blockchain integration and diversification of mathematical foundations to guard against possible weaknesses in lattice-based constructions; the quantum-attack algorithms themselves (Grover, Shor) are only at demonstrator stage.

³⁰ <https://csrc.nist.gov/News/2016/Public-Key-Post-Quantum-Cryptographic-Algorithms>

Annex IV – About the authors

Data Intelligence for Policy

The JRC.T.5 Data Intelligence for Policy of the European Commission specialises in large-scale text data analysis applying AI-driven approaches to provide actionable insights for policy. The unit delivers data, methods, services, and analysis to scientists and policymakers throughout the JRC and EU institutions.

As part of the Unit's activities, the *Political Intelligence* and *Information Integrity* teams aim to inform and enhance policymaking by providing a comprehensive view of the main topics and entities dominating the media landscape relevant to specific policy domains and assessing the associated perception. To achieve this, the teams aggregate and analyse various data sources, including news media collected by [Europe Media Monitor \(EMM\)](#),³¹ political speeches, public opinion, and existing and upcoming legislation. By applying cutting-edge text mining techniques, the teams extract valuable knowledge on key topics, sentiment, framing, and other aspects relevant to informing policymakers.

The Unit's *Technology Innovation Monitoring (TIM)* team provides an insight in the scientific developments, leading actors and technology foresight, by using bibliometric data sources.

Please check https://knowledge4policy.ec.europa.eu/text-mining/topic/political-intelligence_en for more information on Political Intelligence.

European Science-Media Hub

The European Science-Media Hub (ESMH), operating under the political responsibility of the European Parliament Panel for the Future of Science and Technology (STOA), is a platform to promote networking, training, and knowledge sharing between the European Parliament, the scientific community, and the media.

The ESMH creates a network among policy-makers, scientists, and media involving science, academia, educational and research entities, and professional associations of journalists and scientists. For journalists and media representatives, the ESMH organises training sessions and workshops on current technological developments, both as subjects of their reporting and as means of facilitating their work. Via media monitoring and media intelligence tools, the ESMH follows the most popular topics in the field of science and technology on different platforms including journals, newspapers, and social media.

The ESMH makes information available to journalists, other media, and citizens about new scientific developments, as well as about scientific topics that attract media attention, and promotes information based on evidence.

Check <https://sciencemediahub.eu/> for more information, methodology and technology.

³¹ EMM is a tool developed and maintained by the Data Intelligence for Policy Unit. Its main purpose is to monitor a large set of online media, reducing the information flow to manageable proportions by clustering related news, categorising articles and applying language-technology tools to derive further metadata, such as recognising and disambiguating entities in the text, extracting quotes by and about people, and more.

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